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Genome-wide identification of *NHX* gene family and effects of salicylic acid in regulating antioxidant activity as well as nutrients deposition under salt stress in three *Brassica* species

Rana Muhammad Amir Gulzar¹ , Raheel Munir¹, Zhang Ren Yi¹, Nazia Jan¹, Abdul Rauf⁴, Zohaib Kaleem², Muhammad Haseeb Javaid¹, Mumtaz Ali Saand^{3*}, Ali El-Keblawy⁵, Mohamed S. Sheteiwy^{6,7*} and Zaid Ulhassan^{2*}

Abstract

Background The Na^+/H^+ antiporter (*NHX*) gene subfamily is essential for plant adaptation to salt stress, contributing significantly to ion homeostasis. These antiporters play crucial function in various biological processes by regulating Na^+ and H^+ balance within plant cells. Yet their genomic features and roles of salicylic acid (SA) in regulating them, activating antioxidant defense mechanisms as well as nutrients acquisition under salt stress conditions remain largely unknown in many crop species.

Results In this study, we identified the *NHX* gene family members of three Brassicaceae species including *Brassica napus*, *Brassica rapa* and *Brassica oleracea* and analyzed the expression patterns of *NHX* genes in response to salt stress and phytohormonal application (SA). Comprehensive computational analyses, encompassing domain and motif prediction, cellular localization, *cis*-regulatory elements (CREs), and post-translational modification (PTM) sites revealed distinct group-specific characteristics within the *NHX* gene family. These characteristics included chromosomal distribution, motif composition, exon count and protein length (amino acids). qRT-PCR analysis depicted significantly induced expression of *NHXs*, specifically, *BnNHX7*, *BraNHX6* and *BoNHX6* (4 ~ 6 folds) using different concentrations of NaCl (100 mM and 200 mM) and SA (10 mg/L and 20 mg/L) at different periods of time. SA induced antioxidant enzymatic activity (SOD 3 ~ 4 folds, POD 2 ~ 3 folds, CAT 1 ~ 2 folds and APX 11 ~ 12 folds), upregulated antioxidant genes (*SOD*, *POD*, *APX*, *CAT*) by 3 ~ 5 folds under salt stress, and improved nutrient acquisition in all respective Brassicaceae species.

*Correspondence:
Mumtaz Ali Saand
mumtazsaand@gmail.com
Mohamed S. Sheteiwy
msalah@uaeu.ac.ae
Zaid Ulhassan
11616086@zju.edu.cn

Full list of author information is available at the end of the article



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Conclusions We have conducted the first molecular characterization of *NHX* genes in three *Brassica* species, revealing their dual role in salt stress adaptation through ion homeostasis and salicylic acid induced antioxidant regulation, thus identifying promising targets for crop improvement.

Clinical trial number Not applicable.

Keywords Brassicaceae, *NHXs*, CREs, qRT-PCR, Plant stress tolerance, Salicylic acid

Background

Soil salinization poses a significant challenge to sustainable agricultural development. Despite their relative tolerance, *Brassica* crops suffer yield losses under saline conditions in many other plant species [1]. Globally, salinity impacts approximately 800 million hectares of land, with China alone accounting for about 21% of its cultivated area being saline-alkali soil [2, 3]. Brassicaceae have emerged as a valuable model for studying plant responses to biotic and abiotic stresses due to their robust defense mechanisms and metabolic adaptations, which enable survival under extreme climatic conditions [4]. In such a scenario, developing salt-tolerant varieties is the only viable solution. Therefore, elucidating the mechanisms of salt tolerance is crucial for enhancing the genetic improvement of crops.

The *NHX* gene subfamily belongs to the *CPA1* gene family. Na^+/H^+ antiporters are multifunctional membrane transporters that mediate electroneutral $\text{K}^+/\text{Na}^+-\text{H}^+$ exchange, critically regulating cellular pH, ion homeostasis, cell expansion, and salt stress tolerance [5]. It has a Na^+/H^+ exchange (PF00999) protein domain [6]. Moreover, the members of this family may have a similar structure throughout Brassicaceae species, containing 5 to 13 transmembrane domains. In plants, *NHXs* are classified into three main groups including PM-class (*NHX7* and *NHX8*) [7]; Endo-class (*NHX5* and *NHX6*) located in the endosomal compartments and named as Endo-class [8]; and lastly, some *NHXs* are localized in the tonoplast (*NHX1* to *NHX4*) and named as Vac-class [9].

NHXs have been identified in several plant species including *Arabidopsis thaliana* (*AtNHX1-AtNHX8*), grapes (*VvNHX1-VvNHX8*) [10]; 10 *CqNHX* genes in the genome of quinoa (*Chenopodium quinoa* Willd.) [11]; 23, 24, 12, and 12 *NHX* genes were identified in *Gossypium hirsutum* (Gh), *G. barbadense*, *G. arboreum* and *G. raimondii*, respectively [12]; 7 *NHX* genes from honeysuckle [13]; 26 *NHX* in *Cucurbita moschata* [14]; and 7 *NHX* in sorghum [15].

The first *NHX* gene, *NHX1*, was predicted in *A. thaliana* and found responsive to salt stress as *OE* plants of *NHX1* increased salt stress tolerance in *A. thaliana* plants [16, 17]. Total eight *NHX* proteins identified in *A. thaliana*, including *NHX1* and *NHX2*, have been considered the most abundant proteins [18]. *NHX* proteins in *Arabidopsis thaliana* are involved in Na^+ transport from

roots to shoots and stomatal regulation [19, 20]. They also play roles in protein trafficking and enhancing tolerance to salt stress [21, 22]. Likewise, seven *PtNHX* genes, dealt with multiple abiotic stresses in plants counting heat, ABA, drought or cold exposure. Among these identified *NHX* proteins, *NHX7* was found responsive to salt stress through the CBL-CIPK pathway [23]. Moreover, the *NHX* gene, *PeSOS1* was found to be involved positively in salt tolerance in *Populus trichocarpa* [24]. *BvNHX5* is another protein identified in *Beta vulgaris*, proposed to be involved in regulating salt tolerance by interacting with CIPK and CBL proteins [25]. Furthermore, *GbNHX2* in cotton specie, *Gossypium herbaceum*, enhanced significantly high expression in response to high salt stress treatment [26]. Together, these studies illustrated positive impact of identified *NHX* genes in many crop species in enhancing regulation tolerance in plants against salt stress.

Salicylic acid (SA) functions to modulate several physiological processes and protect cells against the detrimental effects including cell death or ion imbalance by acting as a signaling molecule [27]. SA induced several defense and development related mechanisms including antioxidant activity, photosynthetic mechanism and nitrogen metabolism. As it is reported that the concentration of SA varied in different plant species and also can be influenced by several environmental key factors [28]. Similarly, the endogenous SA level is closely linked to different developmental stages of plants as well as external stimuli [29]. In contrast, plants could induce their ability against most of the abiotic stress conditions like drought, heavy metal stresses, high temperature and salinity [30, 31]. As it is reported that plants under abiotic stress conditions may activate several stress responsive genes through the SA-dependent pathways by the synthesis of secondary metabolites and molecular chaperones to enhance stress resilience [32, 33].

Na^+/H^+ antiporter genes are essential for sodium homeostasis in certain plants. Despite evidence of *NHX*-mediated salt tolerance in crops, their genomic structure and response to salt and SA treatments remain uncharacterized in key Brassicaceae species. This study analyzed the evolutionary relationships, gene structures, and CREs of *NHX* genes in three Brassicaceae species, identifying conserved features and potential Na^+/H^+ antiporter functions through predicted phosphorylation

and glycosylation sites. Interaction network analysis suggested that NHX proteins may contribute to salt tolerance synergistically by interacting with other stress-related genes. Transcriptional profiling highlights the regulatory role of *NHX* genes under salt stress and phytohormonal treatment, though underlying mechanisms remain to be fully elucidated. Antioxidant enzyme assays indicated that salicylic acid (SA) mitigates NaCl-induced stress by enhancing the activity and expression of defense-related enzymes, including SOD, POD, CAT, and APX. Nutrient availability to plants has been noticed to improve after the application of SA and found to be reduced after treating plants with NaCl stress. This is the first study that examines SA-induced upregulation of *NHX* genes in *Brassica* species under salt stress, highlighting their roles in antioxidant defense and nutrient homeostasis. The findings offer targets for enhancing salt tolerance in Brassicaceae and suggest broader functions in biotic and abiotic stress responses.

Materials and methods

Identification of NHX proteins and construction of the phylogenetic tree

BLASTP searches were conducted using AtNHX proteins as queries against sequenced green plant genomes in the NCBI database (accessed on July 3, 2024). Retrieved sequences were aligned with AtNHX proteins using ClustalW2 (default settings, accessed on July 10, 2024) and analyzed with GeneDoc [34]. Physicochemical properties, including isoelectric point (pI) and amino acid composition, were assessed using the ProtParam server (accessed on July 18, 2024) [35].

The ClustalW function in MegaX software was used to align AtNHXs with Brassicaceae NHXs (BnNHXs, BraNHXs, BoNHXs). A phylogenetic tree was constructed in MegaX (v10.2.2) [36] using the maximum likelihood (ML) method with 1,000 bootstrap replicates and a 95% site coverage cut-off for gap handling. Brassicaceae NHX family members were named based on sequence similarity to AtNHX proteins in the phylogenetic tree.

Prediction of domains, motifs, gene structure, and subcellular localization

Two online tools, Simple Modular Architecture Research Tool (SMART v9.0) and NCBI's Conserved Domain Database (CDD v3.20) with the E -value $\leq 1 \times 10^{-10}$ as the threshold were assessed to identify the domain architecture in NHX proteins [37, 38]. Secondly, prediction of conserved motifs has been made by using the 'MEME' tool in NHX proteins (accessed on July 29, 2024) with a maximum of 10 motifs, a width of 6–50 residues, and a site distribution of up to one occurrence per sequence. To assess the conservation among the motifs, the WebLogo3 (v3.7) software was used to assign the motif logos [39].

Thirdly, gene structure was analyzed by using GSDS software to examine the CDS and exon/ intron organization of *NHX*s. Lastly, DeepLoc 2.0 was used to predict the subcellular localization of Brassicaceae NHX proteins [40].

Prediction of PTM-sites in BnNHX/ BraNHX/ BoNHX proteins and promoter profiling

The predicted sites of PTMs, including N-glycosylation sites at asparagine (Asn) and phosphorylation sites at threonine, serine, and tyrosine, identified for a better understanding of their functions in plant stress conditions [36]. <http://www.cbs.dtu.dk/services/NetOGlyc/>, accessed on 02 August 2024 (Glycosylation); <http://www.cbs.dtu.dk/services/NetPhos/>, accessed on 02 August 2024 (Phosphorylation). Additionally, promoter analysis was performed to predict the potential *cis*-regulatory elements (CREs) in Brassicaceae *NHX*s, 2 kb of upstream sequences were retrieved from NCBI and then analyzed by using the search scan program in the Plantcare database [35].

Prediction of tertiary protein structure of BnNHX/ BraNHX/ BoNHX

The tertiary structures of NHX proteins were predicted using Phyre², obtaining pdb files of NHX proteins (accessed on July 29, 2024). Transmembrane protein channels were identified from 3D structures using the PoreWalker webserver (accessed on August 2, 2024) with respective PDB files. Secondary structure validation was performed using the Protter online tool (accessed on August 3, 2024) [36].

Proteins interaction network analysis

The protein-protein interaction (PPI) analysis of Brassicaceae NHX proteins was conducted using the *Arabidopsis* model in the STRING database [41].

Gene expression analysis

Expression analysis was conducted on 25 *NHX* genes using plant samples collected at various growth stages, including young leaves, mature leaves, flowers, roots, and stems. One-month-old Brassicaceae leaves were treated with NaCl and SA, and samples were collected at 1, 3, 6, 12, and 24 h post-treatment. Leaves treated with an equivalent volume of distilled water (mock treatment) were used as the control for all gene expression experiments. Extraction of total RNA was done by using Trizol reagent (Vazyme, Nanjing, China) following standard protocols [36]. qRT-PCR experiment was conducted using a StepOne Real-Time PCR system (Applied Biosystems, Waltham, MA, USA) with SYBR Green PCR Master Mix (TaKaRa, Dalian, China). Relative gene expression levels were calculated using the $2^{-\Delta\Delta CT}$

method with two technical replicates and three biological replicates per sample. Statistical significance was assessed using a Student's *t*-test with a fold change of ≥ 2 was used as the threshold to define differentially expressed genes [39]. Actin (*BnActin*: XP_013678548.1; *BraActin*: XP_033138402.1 and *BoActin*: XP_013620685.1) served as the internal or reference control. The specific gene primers used in present study are listed in Table S3.

Enzymatic antioxidants activity

Superoxide dismutase (SOD)

Previously described method [42] was used to analyze the SOD-activity. The reaction mixture, with a total volume of 700 μ L, consisted of 100 μ L of L-methionine, 50 μ L of nitroblue tetrazolium (NBT), 50 μ L of riboflavin, 250 μ L of phosphate buffer, 100 μ L of Triton X-100, and 150 μ L of distilled water. Spectrophotometer was used for measuring the absorbance at 560 nm by illuminated reaction mixture for 20 min to initiate reaction.

Peroxidase (POD)

According to previous method [43], POD-activity was calculated in Brassicaceae plants. A reaction mixture was prepared in a cuvette containing 0.05 mL of sample extract, 7.5 mL of phosphate buffer, 0.1 mL of guaiacol solution (a mixture of 335 μ L of H_2O_2 and 15 μ L of phosphate buffer), and 0.1 mL of H_2O_2 solution (a mixture of 100 μ L of H_2O_2 and 20 μ L of phosphate buffer). Spectrophotometer was used for measuring the absorbance at 470 nm.

Catalase (CAT)

The catalytic activity in Brassicaceae plant species was measured according to the method previously described by Sofo et al., (2015) [44]. 0.25 g of leaf tissue was ground to homogenized in 4–5 ml of phosphate buffer. Then this mixture was centrifuged at 1500 rpm for 20 min and carefully collect the supernatant. The final reaction mixture contained 0.1 ml of the extracted enzyme solution, 1 ml of hydrogen peroxide (H_2O_2), and 1.9 ml of phosphate buffer. To monitor the enzymatic activity at 30 s time intervals, absorbance was measured by using spectrophotometer at 240 nm.

Ascorbate peroxidase (APX)

About 0.5 g of leaf tissue from sample plants including treated and non-treated plants were homogenized to get enzyme extract in an extraction buffer containing 50 mM potassium phosphate buffer (pH, 7.8), 1 mM ascorbate, 2% PVP, 0.05% Triton X-100 and 1 mM EDTA. The final obtained homogenate centrifuged at $10,000 \times g$ for 20 min, then supernatant was collected to measure the APX activity [45].

Plant materials/ growth conditions and treatments

The seeds of three commercial cultivars, Zhongshuang11, Biyu, and Qingan70 of *B. napus*, *B. rapa*, and *B. oleracea* respectively were obtained from the Institute of Seed Science, College of Agriculture and Biotechnology, Hangzhou, Zhejiang Province, China. Plants were cultivated in growth chambers at 22–23 °C under 14 h/10 h (light/dark) photoperiod in 250-mL plastic pots containing a potting media (peat moss: perlite = 4:1) for four weeks. All the plants were fertilized from a commercial fertilizer, “Kang Pu Jin” 20–20 20ofN-P2O5-K2O Mg TE (COMPO Expert GmbH, Krefeld, Germany), and watered as per necessary in growth chamber. After four weeks, plants were subjected to various treatments, including 100 mM NaCl, 200 mM NaCl, 10 mg/L salicylic acid (SA), 20 mg/L SA, as well as combined treatments of 100 mM NaCl with 10 mg/L SA, 100 mM NaCl with 20 mg/L SA, 200 mM NaCl with 10 mg/L SA, and 200 mM NaCl with 20 mg/L SA for three days. After respective treatments, samples were taken and put it in -80°C refrigerator to assess gene expression, nutrients deposition and antioxidant defense mechanisms later [46].

Measurements of nutrient elements level following SA application in salt stress conditions

The concentrations of essential nutrients, including Mn, Zn, Ca, Mg, P, and Fe, in plant tissue samples were quantified using inductively coupled plasma mass spectrometry (ICP-MS) [47].

Statistical analysis

All experiments were conducted using three biological replicates. Statistical analyses were performed to calculate significant differences by using Student's *t*-test, and the results are presented as the mean $\pm$ standard error (SE) using GraphPad Prism 8.0 software [36].

Results

Phylogeny, identification and physicochemical properties of NHXs

The 8 AtNHX protein sequences were used as queries for BLASTp search against *B. napus*, *B. rapa* and *B. oleracea* genome (*Brana_ZS_PB_V1.0*, *Brara_Chiifu_V3.1*, and *Braol_HDEM_V1.0* respectively) at NCBI/ Ensemble Plants database. Domain analysis performed using SMART and the Conserved Domain Database (CDD) identified the Na⁺/H⁺ exchanger domain in 10, 7, and 8 proteins from the genomes of *B. napus*, *B. rapa*, and *B. oleracea*, respectively, validating their classification as NHX proteins. NHX proteins were named based on their phylogenetic positions relative to *Arabidopsis thaliana* NHXs (BnNHX1–BnNHX10; BraNHX1–BraNHX07; BoNHX1–BoNHX08). The identified proteins were classified into two main classes: Class I (Vac-class), which

were predicted to localize in the vacuole, including AtNHX1-AtNHX4 (BnNHX1-BnNHX7; BraNHX1-BraNHX5; BoNHX1-BoNHX6); Class II (Endo-class), which were localized in the endosomal compartments, encompassing AtNHX5-AtNHX6 (BnNHX8-BnNHX10; BraNHX6-BraNHX7; BoNHX7-BoNHX8). There is another class, Class III (PM-class), which were predicted to localize in plasma membrane but did not include any *Brassica* NHX homologs, corresponding to AtNHX7-AtNHX8 (Fig. 1).

Intriguingly, the *NHX* genes contained dominantly 4–22 exons (Table 1). The identification and bioinformatic analyses of *NHX* genes in three *Brassica* species revealed that their chromosomal locations and distributions are conserved across the *Brassica* genus (Table 1).

Firstly, *BnNHX* genes (*BnNHX1–BnNHX10*) were distributed in A02 (*BnNHX2*), A05 (*BnNHX6* and *BnNHX7*), A07 (*BnNHX10*), A09 (*BnNHX3*), C05 (*BnNHX5*), C06 (*BnNHX8* and *BnNHX9*), C07 (*BnNHX1*), C09 (*BnNHX4*). Secondly, *BraNHXs* were dispersed in A02 (*BraNHX1*), A05 (*BraNHX2* and *BraNHX4*), A07 (*BraNHX6* and *BraNHX7*), A09 (*BraNHX5*) and A10 (*BraNHX3*). Lastly, *BoNHXs* were localized on C02 (*BoNHX2*), C05 (*BoNHX4* and *BoNHX6*), C06 (*BoNHX7* and *BoNHX8*), C07 (*BoNHX1*) and C09 (*BoNHX3* and *BoNHX5*) (Table 1). These results described that all of the *NHX* genes have been localized on most of the chromosomes. However, the chromosomes were unevenly distributed throughout the sub-genomes (A and C).

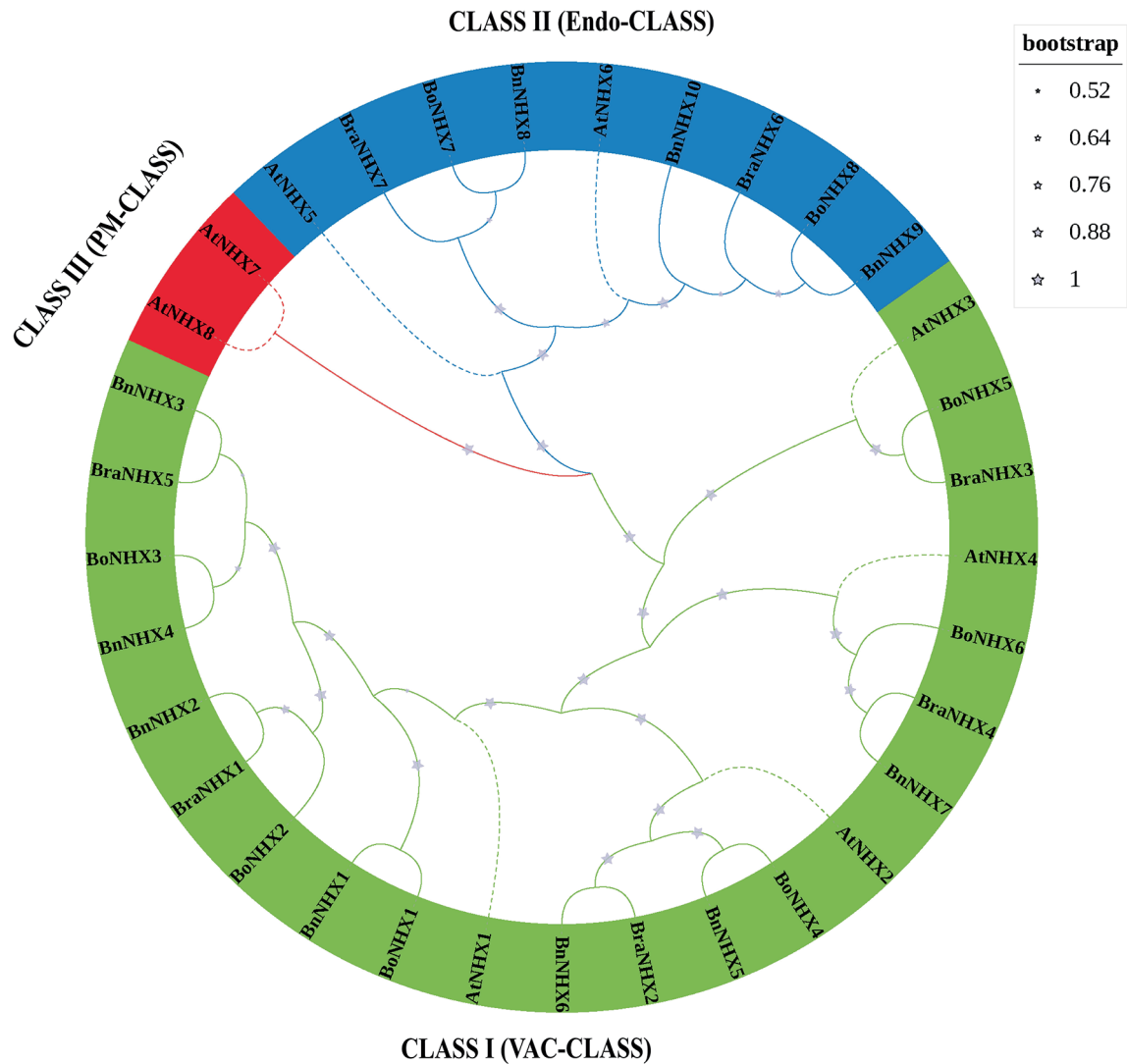


Fig. 1 Phylogenetic tree between 25 Brassicaceae NHX and 8 AtNHX. Maximum likelihood (ML) phylogenetic tree of the proteins of Brassicaceae and *A. thaliana*. A phylogenetic tree was constructed using the MEGA X software with 1000 bootstrap replicates and was optimized with the iTOL online tool. The white filled triangles in the branch indicate bootstrap value. Distinct colors indicate different classes of Brassicaceae, Class I (VAC-Class) NHX in green, Class II (Endo-Class) NHX in Blue, and Class III NHX in Red

Table 1 Characteristics of NHX protein identified in this study. *BnNHX*—*B. napus* Na⁺/H⁺ exchanger (NHX); *BraNHX*—*B. Rapa* Na⁺/H⁺ exchanger (NHX); *BoNHX*— *Brassica oleracea oleracea* Na⁺/H⁺ exchanger (NHX); Chr—chromosome; MW (KDa)— molecular weight (Kilo-Dalton); pI—Iso-electric point; no. of aa—Number of amino acid; GRAVY—Grand average of hydropathicity

Protein ID	Genes	No. of aa	No. of Exon	MW (KDa)	pI	Chromosome	Localization	GRAVY	CDs length (bp)
XP_048620229.1	<i>BnNHX1</i>	445	4	49.7	6.8	ChrC07	Vacuole	0.4	1335
XP_013663082.2	<i>BnNHX2</i>	544	14	60.3	7.2	ChrA02	Vacuole	0.5	1632
XP_022548026.2	<i>BnNHX3</i>	542	14	60.1	7.7	ChrA09	Vacuole	0.5	1626
XP_022562059.1	<i>BnNHX4</i>	535	13	59.3	7.2	ChrC09	Vacuole	0.5	1605
XP_013750682.1	<i>BnNHX5</i>	549	15	61.0	7.1	ChrC05	Vacuole	0.5	1647
XP_013750748.2	<i>BnNHX6</i>	548	15	60.8	7.1	ChrA05	Vacuole	0.5	1644
XP_048636049.1	<i>BnNHX7</i>	558	14	61.8	6.6	ChrA05	Vacuole	0.4	1674
XP_013702788.1	<i>BnNHX8</i>	535	22	59.4	5.3	ChrC06	Endomembrane	0.4	1605
XP_048626893.1	<i>BnNHX9</i>	308	14	34.5	5.9	ChrC06	Endomembrane	0.1	924
XP_013650290.2	<i>BnNHX10</i>	541	22	60.1	5.6	ChrA07	Endomembrane	0.4	1623
XP_009129852.1	<i>BraNHX1</i>	544	14	60.3	7.2	ChrA02	Vacuole	0.5	1632
XP_033147777.1	<i>BraNHX2</i>	548	15	60.8	7.1	ChrA05	Vacuole	0.5	1644
XP_009120057.1	<i>BraNHX3</i>	531	15	58.600	6.9	ChrA10	Vacuole	0.5	1593
XP_009129662.1	<i>BraNHX4</i>	558	14	61.8	6.6	ChrA05	Vacuole	0.4	1674
XP_009111931.1	<i>BraNHX5</i>	543	14	60.2	7.7	ChrA09	Vacuole	0.5	1629
XP_009106685.1	<i>BraNHX6</i>	541	22	60.1	5.6	ChrA07	Endomembrane	0.4	1623
XP_009104514.2	<i>BraNHX7</i>	535	22	59.4	5.3	ChrA07	Endomembrane	0.4	1605
XP_013594973.1	<i>BoNHX1</i>	419	10	46.5	5.8	ChrC07	Vacuole	0.4	1257
XP_013621287.1	<i>BoNHX2</i>	544	14	60.4	7.2	ChrC02	Vacuole	0.5	1632
XP_013611175.1	<i>BoNHX3</i>	535	14	59.3	7.1	ChrC09	Vacuole	0.5	1605
XP_013587125.1	<i>BoNHX4</i>	549	15	60.9	7.1	ChrC05	Vacuole	0.5	1647
XP_013609990.1	<i>BoNHX5</i>	531	15	58.8	7.2	ChrC09	Vacuole	0.5	1593
XP_013583987.1	<i>BoNHX6</i>	560	14	62.2	6.6	ChrC05	Vacuole	0.4	1680
XP_013591702.1	<i>BoNHX7</i>	535	22	59.4	5.2	ChrC06	Endomembrane	0.4	1605
XP_013588440.1	<i>BoNHX8</i>	537	22	59.7	5.7	ChrC06	Endomembrane	0.4	1611

As presented in Table 1, the physicochemical properties of the NHX proteins revealed that their amino acid lengths varied from 308 amino acids (*BnNHX9*) to 560 amino acids (*BoNHX6*), with an average length of 434 amino acids. Isoelectric points (pI) of the NHX proteins ranged from 5.24 (*BoNHX7*) to 7.68 (*BraNHX5*), with an average pI of 6.46. The molecular weights (M.W) of the proteins ranged from 34.5 kDa (*BnNHX9*) to 62.2 kDa (*BoNHX6*), with an average M.W of 48.35 kDa. All NHX proteins were identified as typical transmembrane transporters, containing the Na⁺/H⁺ exchanger domain (Fig. 2A). The number of transmembrane helices in the NHX proteins varied between 5 (*BnNHX9*) and 13 (*BnNHX2/3/4*; *BraNHX1/5*; *BoNHX2/3*). Subcellular localization predictions indicated that most of the NHX proteins are likely localized in vacuoles, while a few (*BnNHX8/9/10*, *BraNHX6/7*, and *BoNHX7/8*) are localized to endomembranes within plant cells.

Gene structure and conserved motif sequence analysis of NHX

To analyze the structure of *NHX* genes, the dispersion of exons and introns were visualized. The intron numbers varied between 3 and 21, with average values of 13.70, 15.57, and 14.75 for *BnNHX*, *BraNHX*, and *BoNHX*,

respectively. This suggests that the *BraNHX* genes have more introns compared to those from other Brassicaceae species (Fig. 2B). Furthermore, significant variation was observed in the size of introns across *NHX* genes of different Brassicaceae species. These findings highlight that the gene structure of *NHX* genes is specie-specific.

To discover the specific amino acid sequence that elaborated the structural diversity of NHX proteins, motif analysis was done for 25 NHX amino acid sequences retrieved from three *Brassicaceae* genomes (*B. napus* (*Zhongshuang11 v1.0*), *B. rapa* (*Biyu v2.0*), and *B. oleracea* (*Qingan70 v2.0*) using the MEME program (V 5.5.7) (with the *E*-value<0.05 as the threshold) (Fig. 3). Total 10 motifs were identified in NHX proteins, and all of the NHX proteins contained maximum of motifs except motif 7. Four *BnNHX* proteins, *BnNHX1/8/9/10* do not contain motif 5, 7, 8/ 7/ 4, 5, 7, 8, 9/ 7 respectively. Moreover, all of the *BraNHX* proteins contained all of the motifs except motif 7 in two *BraNHX* proteins (*BraNHX6* and *BraNHX7*). All *B. oleracea* proteins contained all motifs except *BoNHX1* that is missing of motif 5 and 8; *BoNHX7* and *BoNHX8* do not contain motif 7. As a whole, the motif types and number were not quite different among three Brassicaceae species and proving that NHX proteins were highly conserved among each other.

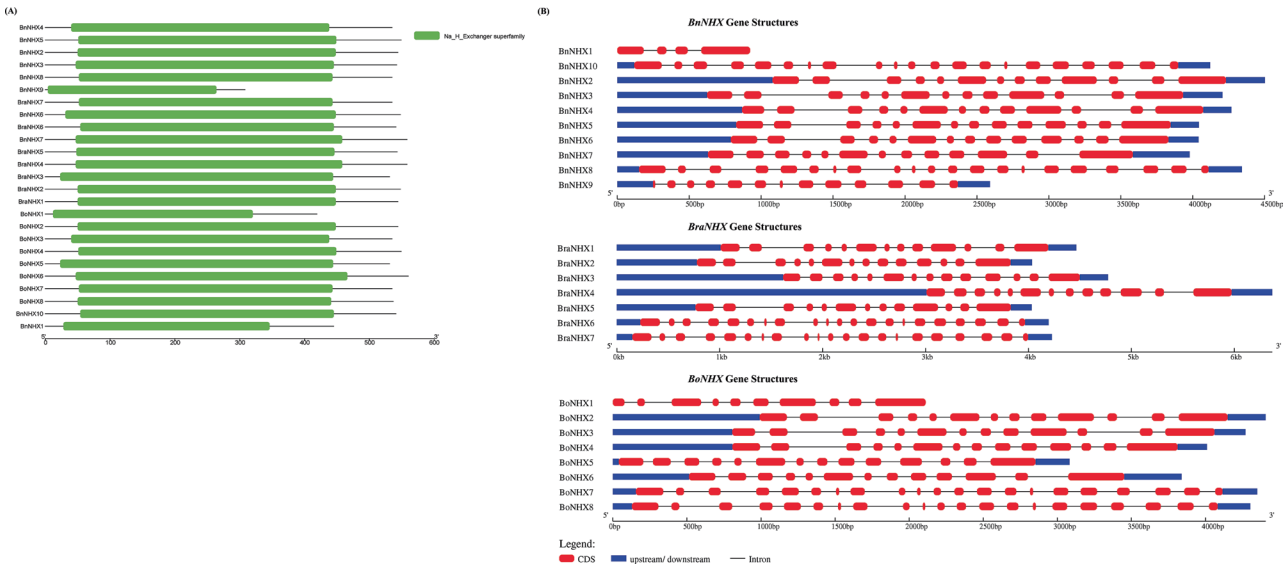


Fig. 2 Conserved domains and the exon–intron structures of 25 Brassicaceae NHXs. **(A)** Conserved domain analysis in the genomes of *B. napus*, *B. rapa* and *B. oleracea* species. **(B)** The constituents of Brassicaceae NHX genes are indicated in differentially colored shapes. CDS: Red box; upstream/ downstream: Blue box; introns: blacklines

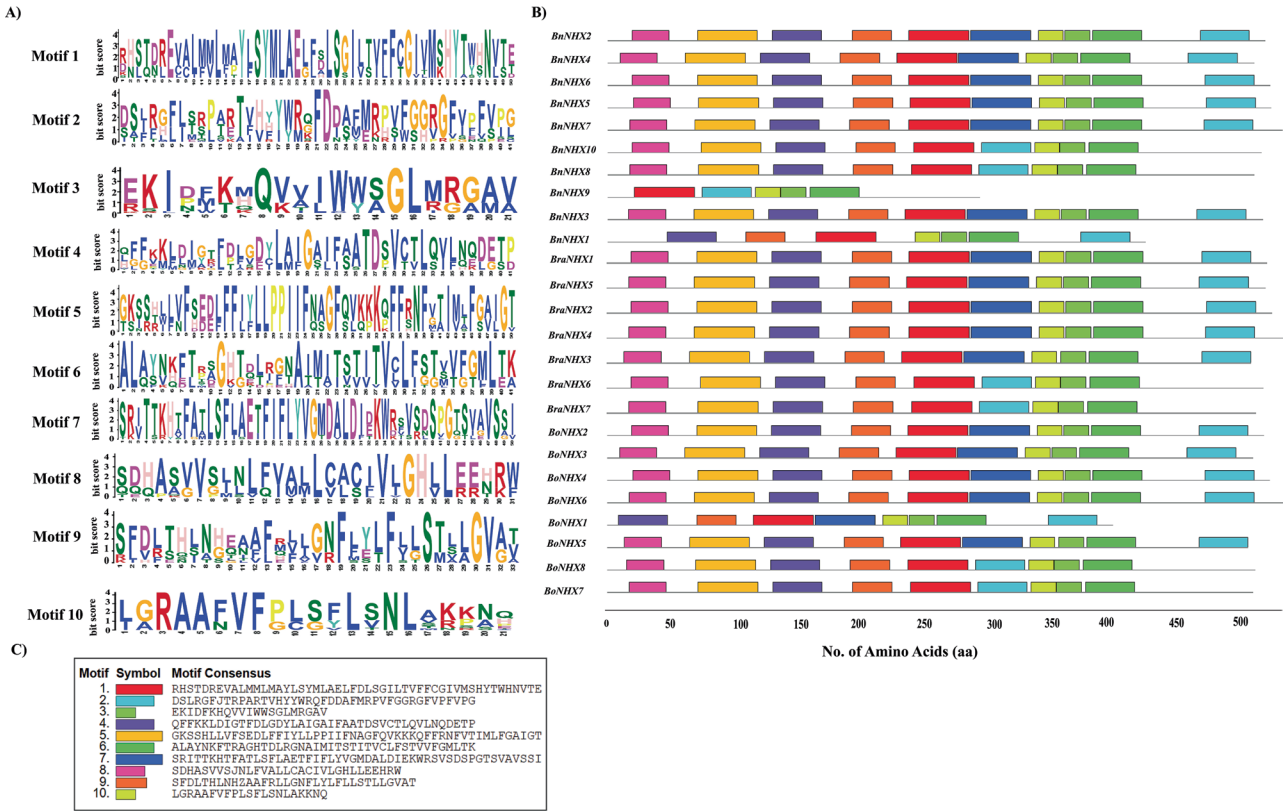


Fig. 3 Conserved motifs among NHX proteins. **(A)** Logos of the predicted motifs. The NHX specific motif logos were generated by using weblogo3 (v3.7). The bit scores for each position in the specific motif logos were generated by using weblogo3. The bit scores for each position in the sequence are indicated to the left. **(B)** Motif profile of Brassicaceae NHXs indicating their position in the proteins. The conserved motifs identified by MEME are indicated by different colors. **(C)** The consensus of each motif is provided

Post translational modification sites in NHX proteins

The phosphorylation of serine, threonine, and tyrosine residues in NHX proteins was predicted. In general, all predicted proteins contained phosphorylation sites for each of these three amino acids. BnNHX10 and BraNHX6 had the highest number of potential serine phosphorylation sites (39), while BoNHX5 had the most threonine sites (27). BnNHX1 exhibited the greatest number (8) of putative tyrosine phosphorylation sites in its protein sequence (Fig. 4A, Table S1). These findings suggest that NHX proteins are likely to be regulated by phosphorylation. Additionally, the phosphorylation levels of protein kinase C (PKC) and UNSP were the highest in NHX proteins, while those of PKG and ATM were comparatively lower (Table S1).

Moreover, on the basis of linkage between amino acid and sugar molecule, glycosylation site can be classified into four main types including N-linked glycans, GPI anchors, C-mannosylation and O-linked glycans.

Analysis of glycosylation patterns in NHX proteins identified O-linked glycans, which is specific for binding of sugar molecule with the hydroxyl group of serine or threonine. Among the NHX proteins, BnNHX7, BraNHX4, and BoNHX6 each contained the highest number of four “Asn” residues, while BnNHX9 had only one “Asn” residue (Fig. 4A, Table S1). On the contrary, all predicted NHXs contained Asn residue implied that all of NHX proteins can also be glycosylated.

Investigation of cis-regulatory elements (CREs) in the promoters of Brassicaceae NHXs

To investigate the potential preliminary functions of NHX genes, CREs in the upstream region ranging 2000 bp of NHX genes were quantified using the search scan program in the Plantcare database. We have found many CREs which can be broadly categorized into light responsive elements (13), stress related elements (7) and plant hormonal responsive cis-elements (6) (Fig. 4B).

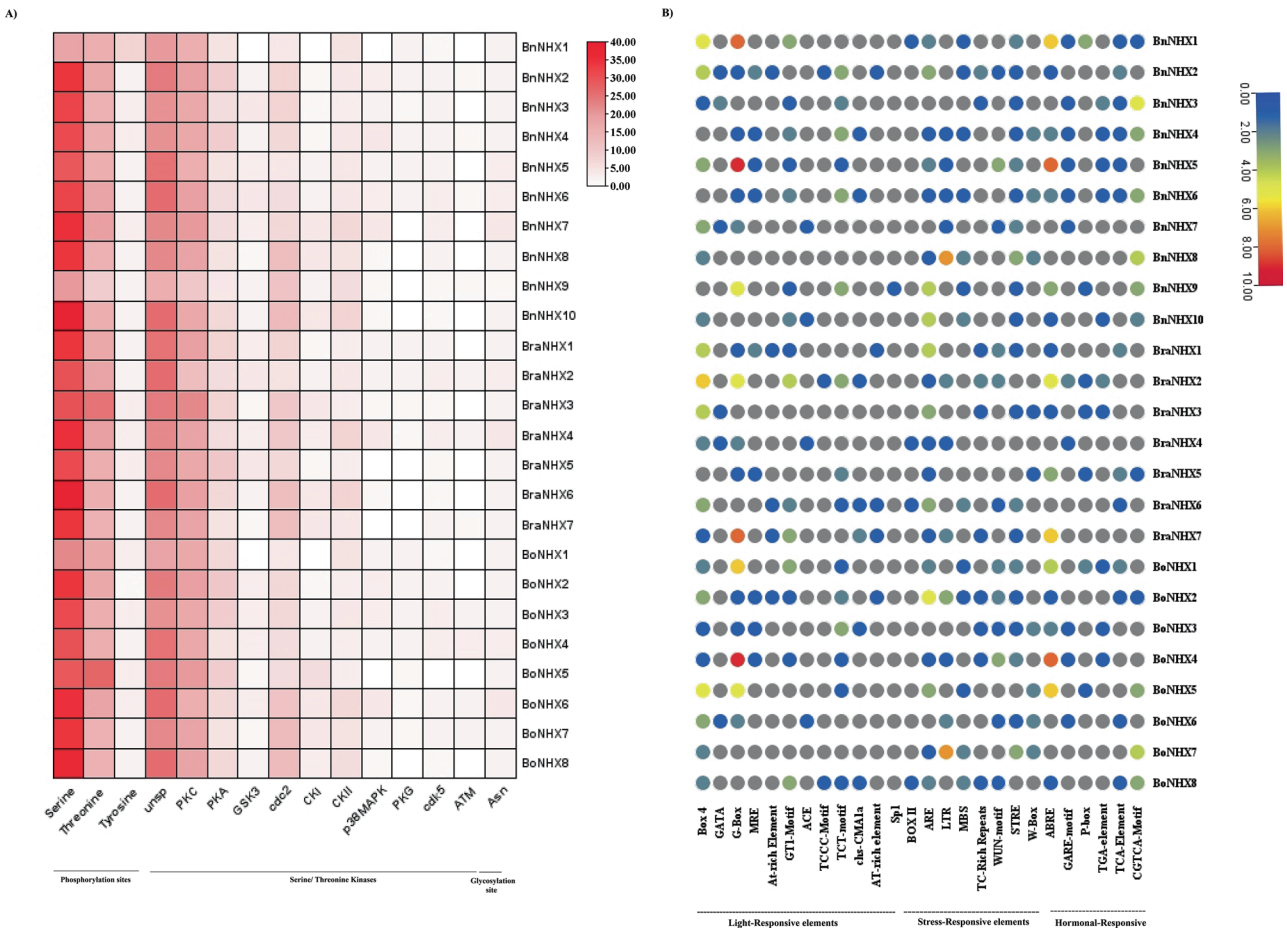


Fig. 4 Predicted phosphorylation and glycosylation sites of NHX proteins and CREs in NHXs' promoters. **(A)** TBtool was used to draw heatmap. Bubble colors (white to red) indicate the number of PTM sites. unsp, unspecified phosphorylation; PKC, protein kinase C; PKA, rotein kinase A; GSK3, glycogen synthase kinase 3; CDC2, cell division cycle protein 2; CKI, casein kinase 1; CKII, casein kinase 2; P38MAPK, P38 mitogen activated protein kinase; PKG, protein kinase G; CDK5, cyclin dependant kinase 5. **(B)** Major stress and hormonal responsive cis-elements predicted in Brassicaceae NHX are shown. The types of cis-elements are provided. The bubble heat map showing the number of elements was prepared using the TB-tool (v2.084)

Most of these genes also contain light responsive elements like Box 4 (6 in *BraNHX2*), G-box (6 in *BnNHX5* and *BoNHX4*), and GT1-motif (4 in *BnNHX2*). Several stress responsive elements were also found in their promoters indicating their important contribution in plant stress responses (biotic/ abiotic). Some of the frequently found stress responsive elements were ARE (5 in *BoNHX2*), LTR (7 in *BnNHX8*), STRE (*BnNHX8* and *BoNHX7* both have 3 *cis*-elements) while TC-Rich Repeats (TC-RR) was found in a very few *NHX* promoters. Some plant hormonal responsive *cis*-elements were also predicted including ABRE elements that were present in almost all of genes, maximum 8 (*BnNHX5* and *BoNHX4*) to minimum 0 (*BnNHX3/7/8*, *BraNHX4/6* and *BoNHX6/7*). It is noteworthy that AT-TATA-box and CAAT-box were found abundantly in most of the genes (145 and 609 respectively) showing their importance in regulating gene functions. Some other CREs were also found in *NHX*-promoters including O₂-site, sp-1 and circadian (Table S2).

Analysis of tertiary protein structure of NHXs proteins

Online webserver, PoreWalker, was used to predict the transmembrane protein channels from their 3D structures. One *NHX* protein from each of Brassicaceae species was used for this analysis including *BnNHX7*, *BraNHX6* and *BoNHX6* which are significantly expressed in qRT-PCR after NaCl and SA application. The pore structure and 3D geometry of these *NHX* proteins have shown a pore morphology that longitudinally passes through the extracellular to intracellular opening of the protein. The pore size and constraints, that are considered as selective barriers and were clearly shown by pore morphology (Fig. 5A-I). In *NHX* proteins, conservation of pore size may lead to propose that these genes might be involved fluxing substances from the cell. For better visualization of the *NHX* proteins' structure, an online tool, Protter, was used. These structures of proteins depicted their functional annotations in Brassicaceae plants.

Protein interaction network analysis

A protein-protein interaction (PPI) network was constructed using the STRING database to investigate the key involvement of Brassicaceae *NHX* proteins through their interactions with other respective proteins based on both known experimental data and predicted interactions. To predict the PPI network for Brassicaceae *NHX* proteins, homologous genes from *Arabidopsis thaliana* were used. As shown in Fig. 6, the homologs AtNHX1, AtNHX3, AtNHX4, and AtNHX6 (corresponding to *BnNHX2*-*BoNHX2*, *BraNHX3*-*BoNHX5*, *BnNHX7*, and *BraNHX6*, respectively) shared several potential interacting proteins, including HKT1 (a sodium transporter involved in salt tolerance) and CBL4 (a calcineurin B-like

protein that regulates intracellular Na⁺ and K⁺ homeostasis, contributing to salt resilience) [48, 49]. Other interacting proteins included AVP1, a vacuolar proton pump that enhances ion retention and contributes to NaCl and drought tolerance [13]; CIPK24, a serine/threonine kinase that interacts with CBL proteins to regulate ion homeostasis and salt tolerance; and PCR12, a protein involved in heavy metal transport and cadmium resistance [50]. These interactions indicated that *NHX* homologs in Brassicaceae may work alongside other salt tolerance-related proteins to help plants adapt to salt stress.

Relative expression level of NHX in different plant tissues

To find the expression of *NHX* in different plant parts via qRT-PCR, 10 *BnNHXs*, 7 *BraNHX* and 8 *BoNHX* were used by using mature/ young leaves, flower, root and stem of *B. napus*, *B. rapa* and *B. oleracea* plants. It was found that all of the *NHX* genes were differentially expressed in different tissues and may be tissue specific in expression behavior. Most of the *NHX* genes significantly expressed in mature leaves in all of three Brassicaceae species. Five genes from *BnNHXs* (*BnNHX2/5/6/7/9*), two genes from *BraNHXs* (*BraNHX1/3*) and three genes from *BoNHXs* (*BoNHX1/5/6*) were significantly induced in mature leaves. Dramatically, most of the *NHX* genes were not highly expressed in other plant tissues except *BnNHX9* and *BoNHX3* that were highly induced in young leaves. These results illustrated that most of the *NHX* genes were expressed in all plant parts although the expression level is very low (less than 1-fold) (Fig. 7A).

Expression profile of NHXs in response to SA and NaCl treatment

The expression profile of 25 *NHX* genes was investigated against SA application and NaCl stress (Fig. 2B). The transcriptional regulation of most of *NHX* genes were not significantly high (less than 1~2 fold) even after combined treatment of NaCl (100 mM) and SA (10 mg/L) (Fig. S1). Several *NHX* genes were showed very high transcriptional regulation after combined treatment of NaCl (200 mM) and SA (20 mg/L). While some of the genes could not show transcriptional induction even after combined application of NaCl (200 mM) and SA (20 mg/L). Two genes from *B. napus* (*BnNHX2* and *BnNHX7*), two genes from *B. rapa* (*BraNHX3* and *BraNHX6*) and lastly two genes from *B. oleracea* (*BoNHX2* and *BoNHX5*) were showing drastically enhanced regulation after the application of NaCl up to 4~5 folds as compared to control or non-treated plants specially after 24 hpt (Fig. 7B). Instead, SA application also induced most of the *NHX* genes except very few that were not regulated or induced even after the application of SA at high concentration. These genes belonged from all of the three species of Brassicaceae including *BnNHX3/4*, *BraNHX1/4* and

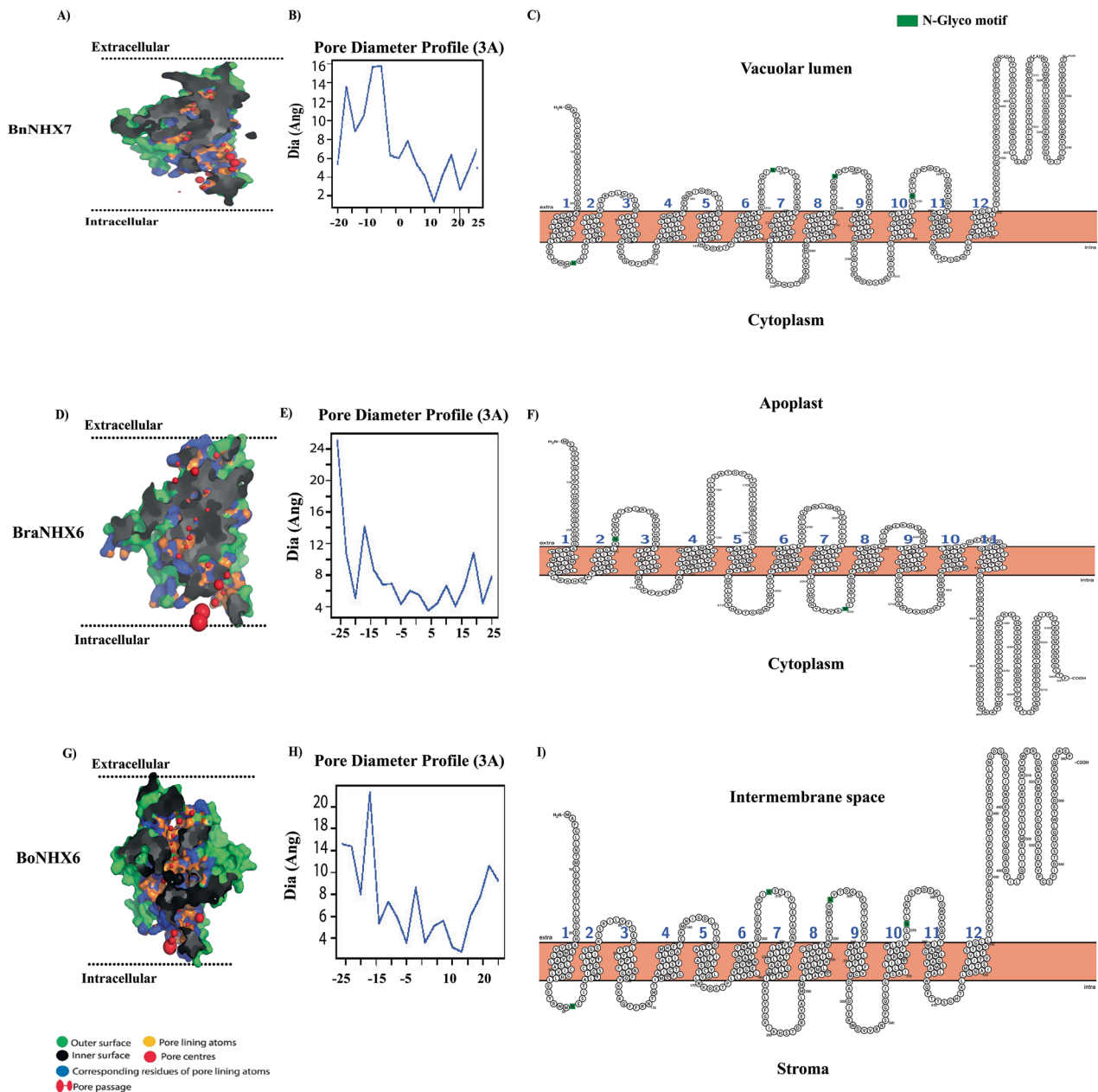


Fig. 5 Pore morphology, dimensions, and topology of Brassicaceae NHX proteins. (A, D, G) Protein tertiary structures of Brassicaceae NHXs. (B, E, H) pore diameter profile of Brassicaceae NHXs. Pore dimensions were obtained using the PoreWalker software (v1.0). (C, F, I) topology of Brassicaceae NHXs. Secondary structure of two of each Brassicaceae NHXs was visualized by the online tool Protter (v1.0). The numbers count the transmembrane domains

BoNHX1/7 (less than 1~2 fold) respectively (Fig. 7B). This expression analysis showed that the response of all *BnNHXs* is specie-specific and may depend on SA or NaCl treatment at different time points.

Impact of NaCl stress on Brassicaceae crops: evaluation of induced resistance

When the SA and NaCl were applied to the *B. napus*, *B. rapa* and *B. oleracea* plants in two different concentrations (10–20 mg/L and 100–200 mM), they remarkably induced

changes in SOD activity when compared to non-treated control plants. The noted SOD activity was at a maximum (486, 547 and 479 IU/mg) after the application of SA (20 mg/L) in *B. napus*, *B. rapa* and *B. oleracea* respectively. Minimum of SOD activity was recorded (136 and 196 IU/mg) in *B. napus* and *B. oleracea* respectively after NaCl treatment (100 mM) as compared to non-treated control plants (Fig. 8A–C). The Brassicaceae plants showed response antagonistically in the case of the POD activity after SA and NaCl application as compared to the control

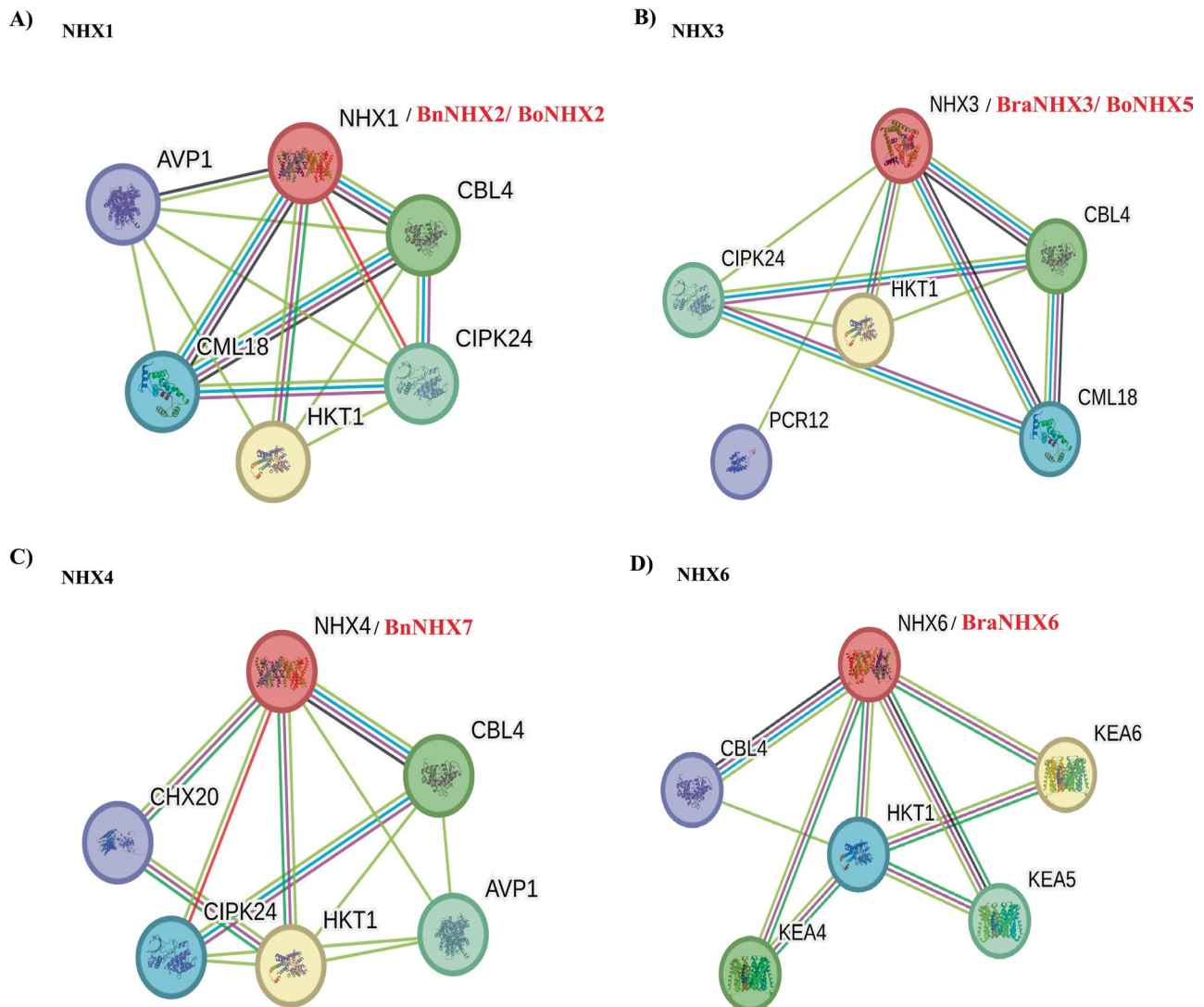


Fig. 6 Prediction of protein-protein interactions (PPIs) between Brassicaceae NHXs and other salt stress-related proteins. **(A)** PPI network among AtNHX1 (homolog of BnNHX2 and BoNHX2) and other stress-responsive proteins. **(B)** PPI network among AtNHX3 (homolog of BraNHX3 and BoNHX5) and other stress-responsive proteins. **(C)** PPI network among AtNHX4 (homolog of BnNHX7) and other stress-responsive proteins. **(D)** PPI network among AtNHX6 (homolog of BraNHX6) and other stress-responsive proteins

plants. The POD activity was found to be the maximum, 3.81, 4.07 and 4.03 IU/mg in *B. napus*, *B. rapa* and *B. oleracea* respectively when SA was applied at 20 mg/L. On the contrary, the minimum POD activity was recorded (1.53, 1.75 and 1.53 in *B. napus*, *B. rapa* and *B. oleracea* respectively) when the NaCl was applied at 100 mM (Fig. 8D–F).

All of the Brassicaceae plants showed maximum catalase activity after the application of SA (20 mg/L) as 2.10, 2.19 and 2.30 in *B. napus*, *B. rapa* and *B. oleracea* respectively and oppositely, minimum activity was found after the treatment of Brassicaceae plants with NaCl (200 mM) as 0.84, 1.11 and 1.13 in *B. napus*, *B. rapa* and *B. oleracea* respectively (Fig. 8G–I). SA regulated the APX activity in the Brassicaceae treated plants as compared to the control plants. Maximum APX activity was recorded (13.14,

14.09 and 15.02 IU/mg in *B. napus*, *B. rapa* and *B. oleracea* respectively) when the SA was applied at 20 mg/L while the lowest APX activity was noticed as 2.16 (at NaCl 100 mM), 2.22 and 2.78 IU/mg (at 200 mM NaCl) was recorded in *B. napus*, *B. rapa* and *B. oleracea* respectively (Fig. 8J–L). Collectively, results concluded that SA could ameliorate enzymatic activity in all of the Brassicaceae plants and ultimately reduce the stress of NaCl in treated plants.

Transcriptional activity of antioxidant related genes in Brassicaceae

We examined the *SOD*, *POD*, *CAT*, *APX* expression levels after treating with all of the Brassicaceae plants with NaCl and SA application. All of the above-mentioned

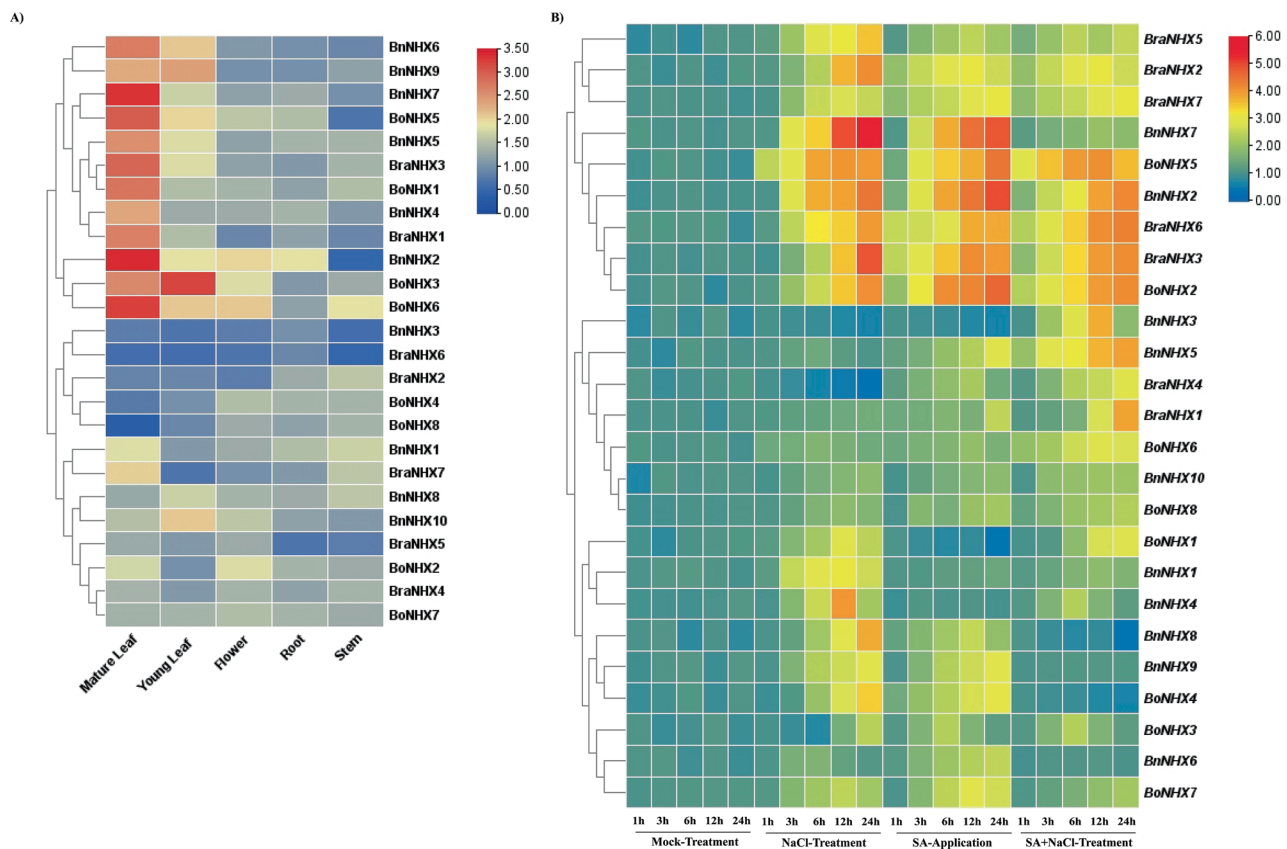


Fig. 7 Transcriptional regulation of Brassicaceae *NHXs*. **(A)** Expression patterns of Brassicaceae *NHXs* in five tissues. All values were expressed relative to the expression levels of reference genes using formula $2^{-\Delta\Delta C_t}$. **(B)** Expression patterns of the Brassicaceae *NHXs* at 1, 3, 6, 12 and 24 h after treated with NaCl (200 mM) and SA (20 mg/L). All values were expressed relative to the expression levels of reference genes using formula $2^{-\Delta\Delta C_t}$ ($n=3$). *Actin* was used as a marker gene for each specific specie. Different colors indicate different levels of gene expression by RT-qPCR. The color from blue to red indicates significant ($P<0.05$) up/down-regulated expression

genes were expressed significantly after NaCl and SA application. *BnSOD*, *BnPOD* and *BnCAT* were induced significantly after the combined application of NaCl (200 mM) and SA (20 mg/L) while *BnAPX* were transcriptionally increased at NaCl (100 mM) and SA (20 mg/L) (Fig. 9A). *BraSOD* and *BraPOD* were significantly expressed (4–5 ~ folds) at highest doses of NaCl and SA (combined application), in contrast *BraCAT* and *BraAPX* were expressed significantly (4 ~ fold) at NaCl (100 mM) and SA (10 mg/L) (Fig. 9B). *BoCAT* and *BoAPX* were shown maximum regulation at NaCl (200 mM) and SA (10 mg/L) whereas *BoSOD* and *BoPOD* at NaCl (100 mM) and SA (20 mg/L) (Fig. 9C). These results demonstrated that all genes involved in antioxidant activity have significant expression showing that SA could accelerate tolerance in Brassicaceae plants under salt stress conditions.

SA could enhance the nutrients deposition in Brassicaceae under salt stress

Salicylic acid (SA), as shown in previous studies, plays a magnificent role in mitigating salt stress. In current study,

SA also considered as potential candidate to enhance nutrient accumulation in plant leaves. Compared to untreated control plants, SA treatment significantly augmented the concentrations of essential nutrients (Fe, Mg, Mn, P and Ca) in leaves of Brassicaceae plants (Fig. 4B). In contrast, NaCl-treated plants exhibited a disruption in ionic homeostasis, with substantial reductions in nutrient levels: Mn (38%), Zn and Mg (10–20%), Ca (20–30%), P (20%), and Fe (15–20%) in the leaves of three Brassicaceae species, relative to the control treatments. This imbalanced nutrient uptake under NaCl stress is likely contributing to the observed stunted growth. The reduced nutrient uptake may be linked to competition between Na^+ and essential ions such as Ca, Mg, and Fe, due to their structural similarities, which disrupts their transport and uptake in plant tissues [51]. However, in plants treated with both NaCl and SA, there was a significant increase in nutrient levels, including Mn (more than 15%), Zn (more than 10%), Ca (more than 20%), Mg (more than 10%), P (more than 15%), and Fe (more than 15%) in the leaves of Brassicaceae species, when compared to controls (Fig. 10).

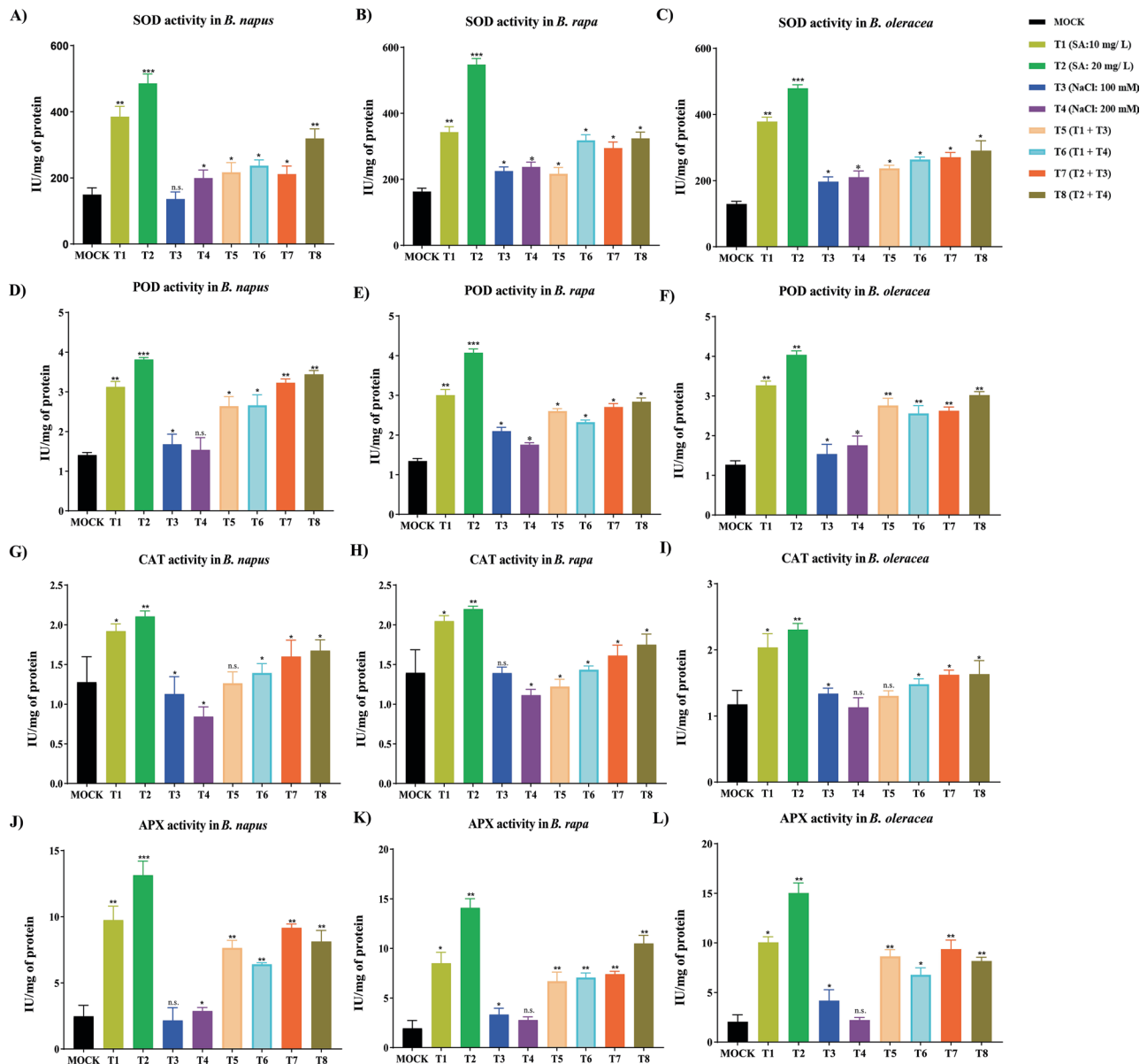


Fig. 8 Effect of SA on antioxidant enzymes in one-month old Brassicaceae species (*B. napus*, *B. rapa*, *B. oleracea*) under salt stress (NaCl). (A–C) Effects of salicylic acid (SA) on superoxide dismutase (SOD) under different salt stresses. (D–F) Impacts of salicylic acid (SA) on Peroxidase activity (POD) under different salt stresses. (G–I) Salicylic acid (SA) induced the catalase activity (CAT) under different salt stress conditions. (J–L) Effects of salicylic acid (SA) on Ascorbate peroxidase activity (APX) under different salt stresses. This experiment was performed with three biological replicates, each yielding similar results. The asterisks indicate significant differences (* $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, n.s.—not significant) in nine different treatments (Mock=Water; T1=SA-10 mg/L; T2=SA-20 mg/L; T3=NaCl-100 mM; T4=NaCl-200 mM; T5=SA-10 mg/L + NaCl-100 mM; T6=SA-10 mg/L + NaCl-200 mM; T7=SA-20 mg/L + NaCl-100 mM; and T8=SA-20 mg/L + NaCl-200 mM) and statistically analyzed by Student's *t*-test

Discussion

In the present study, we identified 10, 7, and 8 *NHX* genes in three Brassicaceae species: *B. napus*, *B. rapa*, and *B. oleracea*, respectively. A number of evidences proved them to be possibly functional *NHXs*. These all *NHX* proteins contained a Na_H domain (Table 1; Fig. 2). Their Na_H domain bore the conserved motif “FLLP-P11” (amiloride binding domain motif) (Fig. 3), which was reported in transmembrane domain III of *NHX* proteins

in *At* plant [18], *S. bicolor* [15], honeysuckle [13], cotton [12], *Cucurbita* L [14]. and Barley [5]. Furthermore, the Brassicaceae *NHX* were predicted to be localized to vacuole or endomembrane and their tertiary structure fitted with membrane channel morphology, as exemplified by BnNHX7, BraNHX6 and BoNHX6 (Fig. 5) as well as claimed for *NHXs* from genus *Oryza* [52]. Nevertheless, further experimental validation is needed to determine whether these *NHXs* function as Na⁺/H⁺ antiporters.

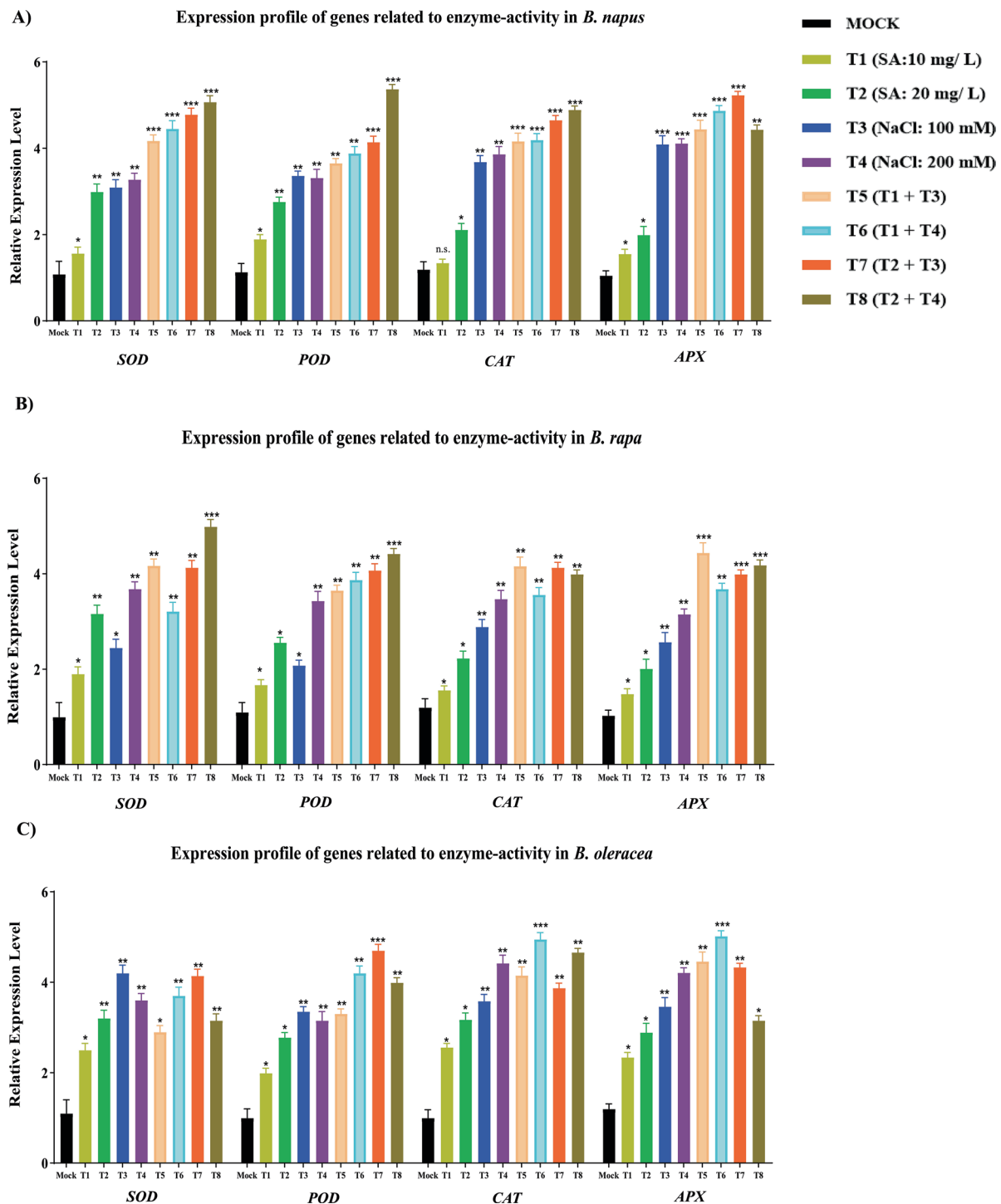


Fig. 9 Transcriptional activation of SOD/ POD/ CAT/ APX genes in response to salicylic acid (SA) application under different concentrations of salt stress. **(A)** Gene expression of four *BnSOD*/*BnPOD*/*BnCAT*/*BnAPX* genes by qRT-PCR, in which *BnActin* gene was used as reference. **(B)** Gene expression of four *BraSOD*/*BraPOD*/*BraCAT*/*BraAPX* genes by qRT-PCR, in which *BraActin* gene was used as reference. **(C)** Gene expression of four *BoSOD*/*BoPOD*/*BoCAT*/*BoAPX* genes by RT-qPCR, in which *BoActin* gene was used as reference. This experiment was performed with three biological replicates, each yielding similar results. The asterisks indicate significant differences (* $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, n.s.—not significant) of antioxidant-related gene expression in nine different treatments (Mock=Water; T1=SA-10 mg/ L; T2=SA-20 mg/L; T3=NaCl-100 mM; T4=NaCl-200 mM; T5=SA-10 mg/ L + NaCl-100 mM; T6=SA-10 mg/ L + NaCl-200 mM; T7=SA-20 mg/ L + NaCl-100 mM; and T8=SA-20 mg/ L + NaCl-200 mM) and statistically analyzed by Student's *t*-test

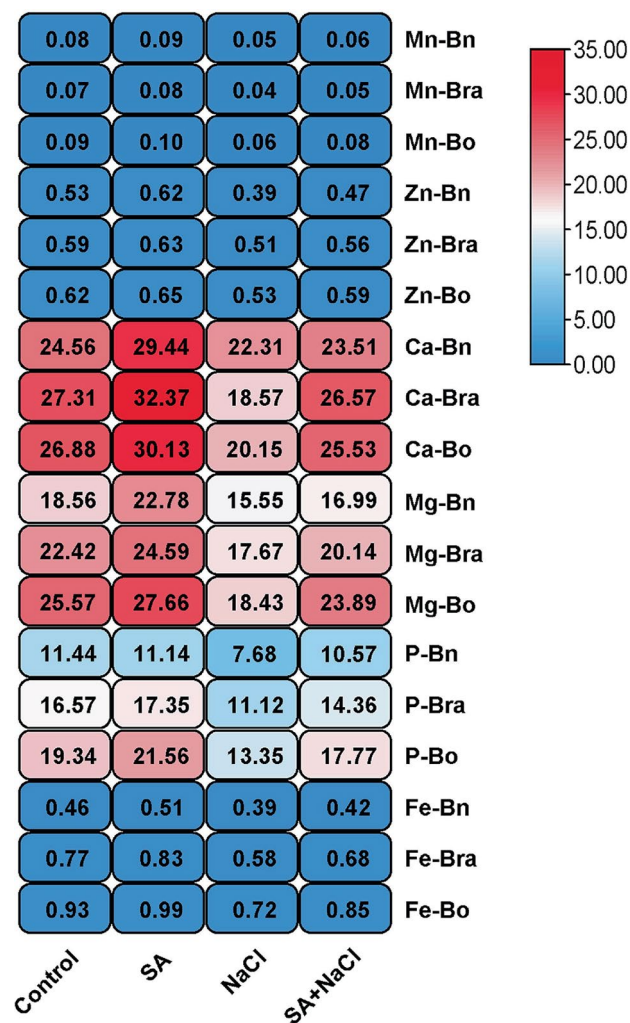


Fig. 10 Effects of SA on nutrients accumulation in leaves under salt stress. TBtool was used to draw heatmap. Bubble colors (Blue to red) indicate the nutrients accumulation level in Brassicaceae plants. Bn, *Brassica napus*; Bra, *Brassica rapa*; Bo, *Brassica oleracea*; Mn, Manganese; Zn, Zinc; Ca, Calcium; Mg, Magnesium; P, Phosphorous; Fe, Iron

In the current investigation, species-specific patterns in *NHX* chromosomal localization were observed. Although the majority of *BnNHX* genes exhibit well-defined chromosomal localization within both the A and C subgenomes of *B. napus*, *BraNHX* genes are notably absent from chromosomes A01, A03, A04, A06, and A08, while *BoNHX* genes are absent from chromosomes C01, C03, C04, C08, and C10 (Table 1). Secondly, the gene structure of *NHXs* does not distinguish specie-wide. Most of the Brassicaceae *NHX* genes contain 12–13 introns while some of them contain maximum 20 introns (*BnNHX8/10*, *BraNHX7* and *BoNHX7/8*) or minimum 3 introns (*BnNHX1*) (Fig. 2). Thirdly, the motif analysis of Brassicaceae differed specie-wide. For instance, all of the *BraNHX* (except *BraNHX6/7*) contained all the motifs including amiloride motif except motif 7. While

BoNHX7/8 don't have motif 7 and *BoNHX1* is missing of motif 5 (amiloride binding motif) and 8 such as absent in *LjNHX4* [13]. Additionally, motif 7 is also absent in *Bn1/8/9/10*. Generally, it is concluded that only motif 7 is found absent in some of the *NHX* proteins while mostly *NHXs* contain all of the motifs (Fig. 3). Fourthly, the distinguished phosphorylation sites were predicted among Brassicaceae species. The maximum phosphorylation sites 308, 233 and 252 were found at Serine (Ser), in *BnNHX* (*BnNHX1* to *BnNHX10*), *BraNHX* (*BraNHX1* to *BraNHX7*) and *BoNHX* (*BoNHX1* to *BoNHX8*) proteins respectively (Fig. 4A, Table S1). Overall, our findings highlighted the variation in *NHX* proteins among different Brassicaceae species, suggesting that their functions and underlying mechanisms might be different across species.

Most of the *NHXs* in all of the Brassicaceae species contained maximum light responsive, stress responsive and hormones responsive CREs in their promoters indicating that these identified *NHXs* might be involved in abiotic plant stresses (Fig. 4B). *BraNHX2* promoters contained a maximum of 20 light-responsive CREs, while *BnNHX8/ BoNHX7* had maximum stress responsive elements (15) and finally *BnNHX1* had 12 hormone responsive CREs. Our predicted results are closely related to previous study [13, 15] in which maximum of *CmoNHX* and honeysuckle *NHX* contained similar kind of CREs. This demonstrates that these *NHXs* might be directly involved in several stress-related functions either biotic or abiotic stresses in Brassicaceae. Secondly, the prediction of interacting partners of *NHX* proteins elaborated that *NHX1/ NHX3/ NHX4* and *NHX6* (*Arabidopsis* homolog of *BnNHX2- BoNHX2/ BraNHX3-BoNHX5/ BnNHX7/ BraNHX6* respectively) can interact with *HKT1* and *CBL4* (Fig. 6) showed that identified Brassicaceae *NHXs* proteins might be involved in regulating salt stress tolerance as above-mentioned interacting proteins are reported as salt-stress tolerance [48, 53].

Considerably, salt stress is a critical environment factor that greatly influence agricultural productivity on a large scale. It initiates various cellular responses, leading to morphological and physiological changes such as ion toxicity, oxidative damage and osmotic stress. These effects contribute to altered plant development, chlorosis, premature senescence, epinasty and browning which can ultimately result in plant death. Reducing the harmful effects of salinity stress on plant growth and development requires maintaining proper ion balance in the cytosol, a process regulated by the activity of antiporters such as *NHX1* and *SOS1*.

Remarkably, *NHX1* or *SOS1* are considered as Na^+/H^+ antiporters, that may play critical function in regulation of Na^+ ions in cytosol to maintain ion homeostasis in plants. Moreover, enhanced expression of both proteins

in response to high salt level has been studied widely [47, 54]. In this study, some of the genes from three different Brassicaceae species, showed influential transcriptional activity to salt stress and SA application suggesting their significant involvement in salt tolerance mechanisms via SA mediated tolerance mechanisms across three Brassicaceae species. Based on qRT-PCR analysis following exogenous SA and NaCl treatments, *BnNHX2/7*, *BraNHX3/6*, and *BoNHX2/5* exhibited high expression at specific time points after NaCl exposure (24 hpt) and SA application (24 hpt), indicating their potential function in salt stress response and SA-mediated regulation (Fig. 7B). Our findings are closely similar with the previous study [12], in which *GhNHX_1* and *GhNHX_2* expression in cotton leaves increased dramatically after 12 hpt of salt application and remained highest from 12 to 48 h. Furthermore, *TaNHX3* expression in wheat roots and leaves augmented within 24 h, and OE lines of wheat established its involvement in enhancing salt resistance [55]. Subsequently, at low concentrations (NaCl: 100 mM and SA: 10 mg/L) the expression of multiple *NHX* genes was regulated, highlighting their involvement in salt stress adaptation (Fig. S1). Similar kind of expression patterns were observed in *BvNHX* genes in sugar beet [25]. The combined treatment of NaCl and SA further raised *NHX* gene expression, reinforcing the contribution of salicylic acid in mitigating salt stress in Brassicaceae crops. This is consistent with previous findings [56], who reported that systemic SA application enhanced salt tolerance in tomato plants.

Furthermore, to check the response of SA in regulating the salt tolerance in Brassicaceae plants against NaCl-stress, we measured antioxidant defense activity that illustrates the elimination of salt toxicity in Brassicaceae after treating them with SA (Fig. 8). Significantly, levels of enzymatic antioxidants (SOD, APX, POD and CAT), increased to salt stress following the SA application (Fig. 8A-L). SA modulates antioxidant enzyme activity, thereby reducing the detrimental effects of ROS in salt stress condition [57, 58]. SA is involved in protecting cellular membranes by reducing the enhanced H_2O_2 levels from oxidative stress. SA also averts lipid peroxidation by scavenging H_2O_2 directly or indirectly by activating antioxidant enzymes [59]. In the present study, it might be due to an adaptive response of plants to alleviate oxidative damage caused by reactive oxygen species (ROS) as previously reported [60]. These results are aligned with the previous finding [61] in which BTH and SA both ameliorates resistance in cucumber plants by regulating enzymatic activities. SOD serves as the primary scavenger of superoxide radicals ($O_2^{\bullet-}$), catalyzing their conversion into oxygen (O_2) and hydrogen peroxide (H_2O_2). Subsequently, antioxidant enzymes such as CAT, POD, and APX facilitate the breakdown of

H_2O_2 into water and molecular oxygen, thereby reducing oxidative stress [62]. The application of salicylic acid on leaves led to a marked increase in CAT, POD, and SOD activity among all Brassicaceae species under both treated and control conditions (Fig. 8A-L). Additionally, qRT-PCR analysis of *NHX* genes linked to antioxidant enzymatic activity further supported these findings, signifying an association with salt stress mitigation. Our findings further support the idea that SA mitigates the adverse effects of NaCl stress through its antioxidant functions, acting both directly to neutralize H_2O_2 and indirectly by stimulating antioxidant enzyme activity [63]. In addition, application of SA on plant leaves improved nutrient accumulation in Brassicaceae plant leaves, aligning with previous reports [64, 65]. Nevertheless, our findings highlight key constraints of *NHX* proteins in Brassicaceae, functional validation through overexpression and mutant lines was limited by the polyploidy of *Brassica* species and resource constraints that to be addressed in future studies. Moreover, the dual role of *NHX* genes in ion regulation and SA-mediated stress responses offers new insights for developing salt-resistant *Brassica* varieties through molecular breeding and biotechnological interventions.

Conclusion

Genome-wide analysis was conducted for three Brassicaceae species (*B. napus* cultivar Zhongshuang11; *B. rapa* cultivar Biyu; *B. oleracea* cultivar Qingan70). A total of 10, 7, and 8 *NHX* genes were identified in *B. napus*, *B. rapa*, and *B. oleracea*, respectively, exhibiting classification patterns similar to those observed in *A. thaliana*. We examined several species-specific traits, including chromosomal distribution, motif analysis and number of exon/ introns in *NHX* proteins. Moreover, the identified *NHXs* considered as potential Na^+/H^+ antiporters because they have certain phosphorylation and glycosylation sites. Protein–protein interaction analysis revealed that *NHX* proteins may enhance salt stress tolerance via interactions with salt-responsive proteins to improve crop health. Regulation of *NHX* genes via qRT-PCR analysis demonstrated that SA could induce the transcriptional activity of these genes under the effect of NaCl stress conditions. In addition, antioxidant assays demonstrated that SA enhances tolerance to salt stress in Brassicaceae plants. Furthermore, nutrients accumulation in Brassicaceae plant leaves after SA application indicated positive impact of SA in plant development. The prediction of functional characterization of *BnNHX7*, *BraNHX6*, and *BoNHX6* provides a genetic foundation for developing salt stress-resilient Brassicaceae crops via genome editing or transgenic overexpression, optimizing salt tolerance and nutrient homeostasis under salt stress.

Abbreviations

CREs	Cis-regulatory elements
PTM	Post-translational modification
Asn	Asparagine
PPI	Protein-protein interaction
NBT	Nitroblue tetrazolium
NHX	Na ⁺ /H ⁺ antiporter
PKC	Protein kinase C
Ser	Serine

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12870-025-06759-5>.

Supplementary Material 1: Table S1. Predicted number of phosphorylation and glycosylation sites in NHX proteins. Table S2. Predicted number of cis-regulatory elements (CREs) in *NHX* promoters.

Supplementary Material 2: Table S3. Primers used in this study.

Supplementary Material 3: Figure S1. Relative expression level of Brassicaceae *NHXs*. (A) Expression patterns of the Brassicaceae *NHXs* at 1, 3, 6, 12 and 24 h after treated with NaCl (100 mM) + SA (10 mg/L) and NaCl (200 mM) + SA (10 mg/L). All values were expressed relative to the expression levels of reference genes using formula $2^{-\Delta\Delta CT}$ ($n = 3$). *Actin* gene were used as a marker gene for each specific specie. Different colors indicate different levels of gene expression by qRT-PCR. The color from blue to red indicates significant ($P < 0.05$) up/down-regulated expression.

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Author contributions

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Data availability

The data described in this article can be freely and openly accessed at National Center for Biotechnology Information (NCBI) database (<https://www.ncbi.nlm.nih.gov/>).

Declarations

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Competing interests

The authors declare no competing interests.

Author details

¹Institute of Vegetable Science, Zhejiang University, Hangzhou 310058, China

²Institute of Crop Science, Ministry of Agriculture and Rural Affairs Key Laboratory of Spectroscopy Sensing, Zhejiang University, Hangzhou 310058, China

³Department of Botany, Shah Abdul Latif University, Khairpur, Sindh 66020, Pakistan

⁴Department of Horticulture, Bahauddin Zakariya University, Punjab, Pakistan

⁵Department of Applied Biology, Faculty of Science, University of Sharjah, Sharjah 27272, United Arab Emirates

⁶Department of Integrative Agriculture, College of Agriculture and Veterinary Medicine, United Arab Emirates University, P.O. Box 15551, Al Ain, Abu Dhabi, United Arab Emirates

⁷Department of Agronomy, Faculty of Agriculture, Mansoura University, Mansoura, Egypt

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References

- Hu Q, Hua W, Yin Y, Zhang X, Liu L, Shi J, Wang H. Rapeseed research and production in China. *Crop J*. 2017;5(2):127–35. <https://doi.org/10.1016/j.cj.2017.06.005>.
- Wang X, Wang J, Liu H, Zou D, Hongwei Z. Influence of natural saline-alkali stress on chlorophyll content and Chloroplast ultrastructure of two contrasting rice (*Oryza sativa* L. japonica) cultivars. *Aust J Crop Sci*. 2013;7:289–92.
- Wani SH, Kumar V, Khare T, Guddimali R, Parveda M, Solymosi K. Engineering salinity tolerance in plants: progress and prospects. *Planta*. 2020;251:76–29. <https://doi.org/10.1007/s00425-020-03366-6>.
- Jahangir M, Abdel FIB, Kim HK, Choi YH, Verpoorte R. Healthy and unhealthy plants: the effect of stress on the metabolism of Brassicaceae. *Environ Exp Bot*. 2009;67:23–33. <https://doi.org/10.1016/j.envexpbot.2009.06.007>.
- Jabeen Z, Irshad F, Hussain N, Han Y, Zhang G. *NHX*-Type Na⁺/H⁺ antiporter gene expression under different salt levels and allelic diversity of *HvNHX* in wild and cultivated barleys. *Front Genet*. 2022;12:809988. <https://doi.org/10.3389/fgene.2021.809988>.
- Ma H, Bao A. Research on Tonoplast Na⁺/H⁺ antiporter progress. *J Anhui Agricultural Sci*. 2020;48(14):4–8. <https://doi.org/10.1007/s00299-020-0254-9>.
- Shi H, Ishitani M, Kim C, Zhu JK. The *Arabidopsis thaliana* salt tolerance gene *SOS1* encodes a putative Na⁺/H⁺ antiporter. *Proc Natl Acad Sci USA*. 2000;97:6896–6901. <https://doi.org/10.1073/pnas.120170197>.
- Bassil E, Ohto MA, Esumi T, Tajima H, Zhu Z, Cagnac O, Belmonte M, Peleg Z, Yamaguchi T, Blumwald E. The *Arabidopsis* intracellular Na⁺/H⁺ antiporters *NHX5* and *NHX6* are endosome associated and necessary for plant growth and development. *Plant Cell*. 2011;23:224–39. <https://doi.org/10.1105/tpc.110.079426>.
- Apse MP, Blumwald E. Na⁺ transport in plants. *FEBS Lett*. 2007;581:2247–54. <https://doi.org/10.1016/j.febslet.2007.04.014>.
- Ayadi M, Martins V, Ben AR, Jbir R, Feki M, Mzid R, Géros H, Aifa S, Hanana M. Genome wide identification, molecular characterization, and gene expression analyses of grapevine *NHX* antiporters suggest their involvement in growth, ripening, seed dormancy, and stress response. *Biochem Genet*. 2020;58(1):102–28. <https://doi.org/10.1007/s10528-019-09930-4>.
- Shihua GS. Identification and expression model of *CqNHX* gene family in Quinoa. *J Yantai Uni: Nat Sci Eng Ed*. 2020;33(3):270–5. <https://doi.org/10.3390/genes16010070>.
- Fu X, Lu Z, Wei H, Zhang J, Yang X, Wu A, Ma L, Kang M, Lu J, Wang HYS. Genome-wide identification and expression analysis of the *NHX* (Sodium/hydrogen Antiporter) gene family in cotton. *Front Genet*. 2020;11:964. <https://doi.org/10.3389/fgene.2020.00964>.
- Huang L, Li Z, Sun C, Yin S, Wang B, Duan T, Liu Y, Li J, Pu G. Genome-wide identification, molecular characterization, and gene expression analyses of honeysuckle *NHX* antiporters suggest their involvement in salt stress adaptation. *PeerJ*. 2022;10:e13214. <https://doi.org/10.7717/peerj.13214>.
- Shen C, Yuan J, Li X, Chen R, Li D, Wang F, Liu X, Li X. Genome-wide identification of *NHX* (Na⁺/H⁺ antiporter) gene family in *Cucurbita* L. and functional analysis of *CmoNHX1* under salt stress. *Front Plant Sci*. 2023;14:1136810. <https://doi.org/10.3389/fpls.2023.1136810>.
- Hima KP, Anil KS, Ramesh K, Sudhakar RP, Nagaraju M, Bhanu PA, Shah T, Henderson A, Srivastava RK, Rajashekar G, Chitkineeni A, Varshney RK, Rathnagiri P, Lakshmi NM, Kavi KPB. Genome-wide identification and analysis of *Arabidopsis* sodium proton antiporter (*NHX*) and human sodium proton exchanger (*NHE*) homologs in *Sorghum bicolor*. *Genes (Basel)*. 2018;9(5):236. <https://doi.org/10.3390/genes9050236>.
- Apse MP, Aharon GS, Snedden WA, Blumwald E. Salt tolerance conferred by overexpression of a vacuolar Na⁺/H⁺ antiport in *Arabidopsis*. *Science*. 1999;285:1256–8. <https://doi.org/10.1126/science.285.5431.1256>.

17. Gaxiola RA, Rao R, Sherman A, Grisafi P, Alper SL, Fink GR. The *Arabidopsis thaliana* proton transporters, *AtNhx1* and *AtNhx2*, can function in cation detoxification in yeast. *Proc Natl Acad Sci USA*. 1999;96:1480–1485. <https://doi.org/10.1073/pnas.96.4.1480>
18. Yokoi S, Quintero FJ, Cubero B, Ruiz MT, Bressan RA, Hasegawa PM, Pardo JM. Differential expression and function of *Arabidopsis thaliana* *NHX* Na⁺/H⁺ antiporters in the salt stress response. *Plant J*. 2002;30(5):529–39. <https://doi.org/10.1046/j.1365-3113x.2002.01309.x>
19. Zhang HX, Blumwald E. Transgenic salt-tolerant tomato plants accumulate salt in foliage but not in fruit. *Nat Biotechnol*. 2001;19:765–8. <https://doi.org/10.1038/90824>
20. Shi H, Quintero FJ, Pardo JM, Zhu JK. The putative plasma membrane Na⁺/H⁺ antiporter SOS1 controls long-distance Na⁺ transport in plants. *Plant Cell*. 2002;14:465–77. <https://doi.org/10.1105/tpc.010371>
21. Barragan V, Leidi EO, Andres Z, Rubio L, Luca AD, Fernandez JA, Cubero B, Pardo JM. Ion exchangers *NHX1* and *NHX2* mediate active potassium uptake into vacuoles to regulate cell turgor and stomatal function in *Arabidopsis*. *Plant Cell*. 2012;24:1127–42. <https://doi.org/10.1105/tpc.111.095273>
22. Wu XX, Li J, Wu XD, Liu Q, Wang ZK, Liu SS, Li SN, Ma YL, Sun J, Zhao L. Ectopic expression of *Arabidopsis thaliana* Na⁺(K⁺)/H⁺ antiporter gene, *AtNHX5*, enhances soybean salt tolerance. *Genet Mol Res*. 2016;15:1–12. <https://doi.org/10.4238/gmr.15027483>
23. Tian FX, Chang EM, Li Y, Sun P, Hu JJ, Zhang J. Expression and integrated network analyses revealed functional divergence of *NHX* type Na⁺/H⁺ exchanger genes in Poplar. *Sci Rep*. 2017;7(1):2607. <https://doi.org/10.1038/s41598-017-02607-1>
24. Meng K, Wu YX. Footprints of divergent evolution in two Na⁺/H⁺ type antiporter gene families (*NHX* and *SOS1*) in the genus populus. *Tree Physiol*. 2018;38(6):813–24. <https://doi.org/10.1093/treephys/tpx173>
25. Wu GQ, Wang JL, Li SJ. Genome-wide identification of Na⁺/H⁺ antiporter (*NHX*) genes in sugar beet (*Beta vulgaris* L.) and their regulated expression under salt stress. *Genes*. 2019;10(5):401. <https://doi.org/10.3390/genes10050401>
26. Akram U, Song YH, Li CZ, Abid MA, Askari M, Myat AA. Genome-wide characterization and expression analysis of *NHX* gene family under salinity stress in *Gossypium barbadense* and its comparison with *Gossypium hirsutum*. *Genes*. 2020;11(7):803. <https://doi.org/10.3390/genes11070803>
27. Khan MI, Fatma M, Per TS, Anjum NA, Khan NA. Salicylic acid-induced abiotic stress tolerance and underlying mechanisms in plants. *Front Plant Sci*. 2015;6:462. <https://doi.org/10.3389/fpls.2015.00462>
28. Khan N, Bano A. Effects of exogenously applied Salicylic acid and Putrescine alone and in combination with rhizobacteria on the phyto remediation of heavy metals and Chickpea growth in sandy soil. *Int J Phytorem*. 2018;20(5):405–14. <https://doi.org/10.1080/15226514.2017.1381940>
29. Chavoushi M, Najafi F, Salimi A, Angaji SA. Effect of Salicylic acid and sodium Nitroprusside on growth parameters, photosynthetic pigments and secondary metabolites of safflower under drought stress. *Sci Hortic*. 2020;259. <https://doi.org/10.1016/j.scienta.2019.108823>
30. Huang W, Wang Y, Li X, Zhang Y. Biosynthesis and regulation of Salicylic acid and N-hydroxyphenylpyruvic acid in plant immunity. *Mol Plant*. 2020;13(1):31–41. <https://doi.org/10.1016/j.molp.2019.12.008>
31. Yang W, Zhou Z, Chu Z. Emerging roles of Salicylic acid in plant saline stress tolerance. *Int J Mol Sci*. 2023;24(4):3388. <https://doi.org/10.3390/ijms24043388>
32. Aazami MA, Maleki M, Rasouli M, Gohari G. Protective effects of Chitosan based Salicylic acid nanocomposite (CS-SA NCs) in grape (*Vitis vinifera* Cv. 'Sultana') under salinity stress. *Sci Rep*. 2023;13(1):883. <https://doi.org/10.1038/s41598-023-27618-z>
33. Suzuki N, Rivero RM, Shulaev V, Blumwald E, Mittler R. Abiotic and biotic stress combinations. *New Phytol*. 2014;203:32–43. <https://doi.org/10.1111/nph.12797>
34. Saand MA, Xu YP, Li W, Wang JP, Cai XZ. Cyclic nucleotide gated channel gene family in tomato: Genome-wide identification and functional analyses in disease resistance. *Front Plant Sci*. 2015;6:303. <https://doi.org/10.3389/fpls.2015.00303>
35. Roy BC, Shukla N, Gachhui R, Mukherjee A. Genome-wide analysis of glutamate receptor gene family in allopolyploid *Brassica napus* and its diploid progenitors. *Genetica*. 2023;151:293–310. <https://doi.org/10.1007/s10709-023-00192-y>
36. Gulzar RMA, Ren CX, Fang X, Xu YP, Saand MA, Cai XZ. Glutamate receptor-like (GLR) family in *Brassica napus*: genome wide identification and functional analysis in resistance to *Sclerotinia sclerotiorum*. *Int J Mol Sci*. 2024;25:5670. <https://doi.org/10.3390/ijms25115670>
37. Letunic I, Bork P. 20 Years of the SMART protein domain annotation resource. *Nucleic Acids Res*. 2017;46:D493–6. <https://doi.org/10.1093/nar/gkx922>
38. Wang J, Chitsaz F, Derbyshire MK, Gonzales NR, Gwadz M, Lu S, Marchler GH, Song JS, Thanki N, Yamashita RA. The conserved domain database in 2023. *Nucleic Acids Res*. 2023;51:D384–8. <https://doi.org/10.1093/nar/gkac1096>
39. He YH, Zhang ZR, Xu YP, Chen SY, Cai XZ. Genome-wide identification of rapid alkalization factor family in *Brassica napus* and functional analysis of *BnRALF10* in immunity to *Sclerotinia sclerotiorum*. *Front Plant Sci*. 2022;13:877404. <https://doi.org/10.3389/fpls.2022.877404>
40. Thumhuri V, Almagro AJ, Johansen AR, Nielsen H, Winther O. DeepLoc 2.0: multi-label subcellular localization prediction using protein language models. *Nucleic Acids Res*. 2022;50(W1):W228–34. <https://doi.org/10.1093/nar/gkac278>
41. Szklarczyk DG, Annika LL, David J, Alexander W, Stefan HC, Jaime S, Milan D, Nadezhda TM, John HB, Peer J, Lars JM. STRING v11: protein-protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucl Acid Res*. 2019;47(D1):D607–13. <https://doi.org/10.1093/nar/gky1131>
42. Giannopolitis CN, Ries SK. Superoxide dismutases: occurrence in higher plants. *Plant Physiol*. 1977;59(2):309–14. <https://doi.org/10.1104/pp.59.2.309>
43. Chance B, Maehly AC. Assay of catalase and peroxidase. *Method Enzymol*. 1955;2:764–75. <https://doi.org/10.1002/9780470110171.ch14>
44. Sofo A, Scopa A, Nuzzaci M, Vitti A. Ascorbate peroxidase and catalase activities and their genetic regulation in plants subjected to drought and salinity stresses. *Int J Mol Sci*. 2015;16(6):13561–78. <https://doi.org/10.3390/ijms160613561>
45. Munir R, Yasin MU, Afzal M, Jan M, Muhammad S, Jan N, Nana C, Munir F, Iqbal H, Tawab F, Gan Y. Melatonin alleviated cadmium accumulation and toxicity by modulating phytohormonal balance and antioxidant metabolism in rice. *Chemosphere*. 2024;346:140590. <https://doi.org/10.1016/j.chemosphere.2023.140590>
46. Ilyas M, Maqsood MF, Shahbaz M, Zulfiqar U, Ahmad K, Naz N, Ali MF, Ahmad M, Ali Q, Yong JWH, Ali HM. Alleviating salinity stress in Canola (*Brassica napus* L.) through exogenous application of Salicylic acid. *BMC Plant Biol*. 2024;24(1):611. <https://doi.org/10.1186/s12870-024-05314-y>
47. Ulhassan Z, Ali S, Kaleem Z, Shahbaz H, He D, Khan AR, Salam A, Hamid Y, Sheteiwy MS, Zhou W, Huang Q. Effects of Nanosilica priming on rapeseed (*Brassica napus*) tolerance to cadmium and arsenic stress by regulating cellular metabolism and antioxidant defense. *J Agri Food Chem*. 2025;73:4518–33. <https://doi.org/10.1021/acs.jafc.4c01111>
48. Zhang WD, Wang P, Bao Z, MaQ DLJ, Bao AK, Zhang JL, Wang SM. SOS1, HKT1;5, and NHX1 synergistically modulate Na⁺ homeostasis in the halophytic grass *Puccinellia tenuiflora*. *Front Plant Sci*. 2017;8(1237):576. <https://doi.org/10.3389/fpls.2017.00576>
49. Dong J, Liu C, Wang Y, Zhao Y, Ge D, Yuan Z. Genome-wide identification of the *NHX* gene family in *Punica granatum* L. and their expression patterns under salt stress. *Agronomy*. 2021;11(2):263. <https://doi.org/10.3390/agronomy11020263>
50. Liu Y, Kong L, Gong C, Yang G, Xu E, Chen W, Zhang W, Chen X. Identification of plant cadmium resistance gene family in *Brassica napus* and functional analysis of *BnPCR10.1* involved in cadmium and copper tolerance. *Plant Physiol Biochem*. 2023;202:107989. <https://doi.org/10.1016/j.plaphy.2023.107989>
51. Handa N, Kohli SK, Sharma A, Thukral AK, Bhardwaj R, AbdAllah EF, Alqarawi AA, Ahmad P. Selenium modulates dynamics of antioxidative defense expression, photosynthetic attributes and secondary metabolites to mitigate chromium toxicity in *Brassica juncea* L. *Plants environ*. Exp Bot. 2019;161:180–92. <https://doi.org/10.1016/j.envexpbot.2018.11.009>
52. Solis CA, Yong MT, Zhou M, Venkataraman G, Shabala L, Holford P, Shabala S, Chen ZH. Evolutionary significance of *NHX* family and *NHX1* in salinity stress adaptation in the genus *Oryza*. *Int J Mol Sci*. 2022;23(4):2092. <https://doi.org/10.3390/ijms23042092>
53. Surekha KA, Kangmin K, Manu A, Xinmiao F, Alex H, Jian KZ. The plasma membrane Na⁺/H⁺ antiporter SOS1 interacts with RCD1 and functions in oxidative stress tolerance in *Arabidopsis*. *Pedia Acute-onset Neuropsychol Syndr*. 2006;103(49):18816–21. <https://doi.org/10.1073/pnas.0604711103>
54. Ali A, Raddatz N, Pardo JM, Yun DJ. HKT sodium and potassium transporters in *Arabidopsis thaliana* and related halophyte species. *Physiol Plant*. 2021;171(4):546–58. <https://doi.org/10.1111/plp.13166>
55. Lu W, Guo C, Li X, Duan W, Ma C, Zhao M, Gu J, Du X, Liu Z, Xiao K. Overexpression of TaNHX3, a vacuolar Na⁺/H⁺ antiporter gene in wheat, enhances

- salt stress tolerance in tobacco by improving related physiological processes. *Plant Physiol Biochem.* 2014;76:17–28. <https://doi.org/10.1016/j.plaphy.2013.12.013>.
56. Rao YR, Ansari MW, Sahoo RK, Wattal RK, Tuteja N, Kumar VR. Salicylic acid modulates *ACS*, *NHX1*, *sos1* and *HKT1;2* expression to regulate ethylene over-production and Na^+ ions toxicity that leads to improved physiological status and enhanced salinity stress tolerance in tomato plants Cv. Pusa Ruby Plant Signal Behav. 2021;16(11):1950888. <https://doi.org/10.1080/15592324.2021.1950888>.
57. Poór P. Effects of Salicylic acid on the metabolism of mitochondrial reactive oxygen species in plants. *Biomol.* 2020;10(2):341. <https://doi.org/10.3390/biom10020341>.
58. Elbeltagi HS, Alotaibi HH, Parmar A, Ramadan KM, Lobato AKDS, Elmogy MM. Application of potassium humate and Salicylic acid to mitigate salinity stress of common bean. *Life.* 2023;13(2):448. <https://doi.org/10.3390/life13020448>.
59. Talaat NB, Mahmoud AWM, Hanafy AM. Co-application of Salicylic acid and spermine alleviates salt stress toxicity in wheat: growth, nutrient acquisition, osmolytes accumulation, and antioxidant response. *Acta Physiol Plant.* 2023;45(1):1. <https://doi.org/10.1007/s11738-022-03485-5>.
60. Sarker U, Hossain MM, Oba S. Nutritional and antioxidant components and antioxidant capacity in green morph Amaranthus leafy vegetable. *Sci Rep.* 2020;10(1):1336. <https://doi.org/10.1038/s41598-020-57687-3>.
61. Gulzar RMA, Rehman AU, Umar UUD, Shahid M, Khan MF. Evaluation of genetic and induced resistance phenomena in cucumbers against the root-knot nematode (*Meloidogyne incognita*). *Plant Protect Sci.* 2022;58:338–50. <https://doi.org/10.17221/130/2021-PPS>.
62. Rezayian M, Zarinkamar F. Nitric oxide, calmodulin and calcium protein kinase interactions in the response of *Brassica napus* to salinity stress. *Plant Biol.* 2023;25(3):411–9. <https://doi.org/10.1111/plb.13511>.
63. Roşca M, Mihalache G, Stoleru V. Tomato responses to salinity stress: from morphological traits to genetic changes. *Front Plant Sci.* 2023;14:1118383. <https://doi.org/10.3389/fpls.2023.1118383>.
64. He D, Kaleem Z, Ali S, Shahbaz H, Zhang K, Li J, Sheteiwy MS, Ulhassan Z, Zhou W. Impact of iron oxide nanoparticles on cadmium toxicity mitigation in *Brassica napus*. *Plant Physiol Biochem.* 2025;220:109500. <https://doi.org/10.1016/j.plaphy.2025.109500>.
65. Muhammad S, Ulhassan Z, Munir R, Yasin MU, Islam F, Zhang K, Chen W, Jan M, Afzal M, Muhammad A, Hannan F, Zhou W. Nanosilica and Salicylic acid synergistically regulate cadmium toxicity in rice. *Environ Pollut.* 2025;364:125331.

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